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ABSTRACT

Significant knowledge gaps remain in our understanding of turbulence–cloud–aerosol interactions in the Earth’s atmosphere, and direct numerical simulation (DNS) has increasingly become an indispensable tool to fill such critical knowledge gaps. This study is an extension of our previous DNS model [Gao *et al.*, J. Geophys. Res.: Atmos., **123**(4), 2194–2214 (2018)], with a focus on the activation of aerosol particles into cloud droplets and deactivation of cloud droplets into aerosol particles in a microscale cloud environment. The effects of turbulence intensity, particle curvature, and solute, as well as the initial distributions of the aerosol particles (monodisperse or polydisperse) are investigated. The governing equations for the flow of air, temperature, and water vapor mixing ratio are solved numerically in the Eulerian fashion, assuming homogeneous and isotropic turbulence. The dynamics of the aerosol and cloud particles are calculated with the Lagrangian particle tracking method. The results show that the deviations of the thermodynamic variables from their respective means are significantly reduced, the activation process appears to be delayed, and the deactivation process occurs more rapidly, as the turbulence intensity is increased. The inclusion of particle curvature and solute effects, as well as polydispersity, tends to retard the activation of aerosols into cloud droplets. It is also observed that fluctuations in supersaturation broaden the spread of particle radii, and the broadening is followed by a narrowing as turbulent homogenization reduces thermodynamic fluctuations over time.

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NOMENCLATURE

c_p, C_d	specific heat of air (J/(kg.K)), rate of condensation or evaporation (1/s)	m_l, m_a	mass of liquid water per grid cell (kg), mass of air per grid cell (kg)
$Da, \delta_{k,k_f}, \epsilon_{in}$	Damkohler number (-), delta function (-), energy injection rate (m^2/s^3)	PDF, TKE	probability density function (-), turbulent kinetic energy (m^2/s^2)
$F_b, F_c, \hat{f}_c$	buoyancy force (m/s^2), physical-space, Fourier-space forcing (m/s^2) per unit mass	p_{sat}, q_c	saturation water vapor pressure (N/m^2), instantaneous liquid water mixing ratio (g/kg)
$\mathbf{g}, \mathbf{k}_f$	acceleration vector due to gravity (m/s^2), forcing wavenumber vector (1/m)	$q_v, q_{v,s}$	instantaneous water vapor mixing ratio (g/kg), saturation vapor mixing ratio (g/kg)
k_{min}, k_{max}	minimum wavenumber (1/m), maximum wavenumber (1/m)	r_c, R_v	critical radius (μm), specific gas constant for water vapor (J/(kg.K))
L_h, λ, l	specific latent heat (J/kg), Taylor microscale (m), integral length scale (m)	r_d, r	dry aerosol radius (μm), wet radius of aerosol particle or cloud droplet (μm)
M_l, M_a	molar mass of water (kg/mol), molar mass of air (kg/mol)	$R, R_{w'w'}$	universal gas constant (J/(kg.K)), autocorrelation coefficient (-)
		S_c, σ_l	critical equilibrium supersaturation (-), surface tension of water (N/m)

S_e, S_k	environmental supersaturation (-), particle equilibrium supersaturation (-)
T, τ_p	instantaneous temperature (K), particle response time (s)
$\mathbf{u}, u_{rms}$	instantaneous fluid velocity vector (m/s), root-mean-square velocity (m/s)
$\hat{\mathbf{u}}, \mathbf{V}, \mathbf{X}$	Fourier velocity field (m/s), particle velocity vector (m/s), particle position vector (m)
α_T, α_c	thermal accommodation coefficient (-), mass accommodation coefficient (-)
ϵ, η	turbulent kinetic energy dissipation rate (m^2/s^3), Kolmogorov length scale (m)
κ, k_T	hygroscopicity of solute (-), thermal conductivity in air ($\text{W}/(\text{m}\cdot\text{K})$)
μ_T, μ_v	molecular diffusivities (m^2/s) of temperature, water vapor
ν, p	kinematic viscosity of air (m^2/s), instantaneous fluid pressure (N/m^2)
ρ_a, ρ_l	density of air (kg/m^3), density of water (kg/m^3)
τ_t, τ_c	turbulent mixing timescale (s), microphysical timescale (s)

I. INTRODUCTION

In the Earth's atmosphere, aerosol particles grow and activate into cloud droplets in a supersaturated environment, while cloud droplets evaporate and deactivate into aerosol particles in a subsaturated environment. The growth of small particles and thus activation/deactivation rely on the competition between the curvature effect or Kelvin effect, and the solute effect or Raoult's law.¹ A spherical cloud droplet has a surface that is curved and consists of numerous water molecules. The greater the curvature, the greater the likelihood that water molecules will escape from the surface. This is the curvature effect, which favors evaporation compared to a flat surface. When solutes or dry aerosols are present, they occupy sites on the particle surface that would otherwise be occupied by water molecules. This is the solute effect² which prevents evaporation. Many studies have investigated the role of curvature and solute on activation and deactivation. For example, Yang *et al.*³ studied the condensational growth of cloud droplets formed on polydisperse aerosol particles with the inclusion of curvature and solute effects, and Yang *et al.*⁴ applied large-eddy simulation (LES) and Lagrangian microphysics to investigate activation and deactivation. They reported that the growth of haze or unactivated aerosol particles and cloud droplets have strong impact on cloud properties when supersaturation fluctuations are comparable to mean supersaturation. However, the LES approach does not resolve all scales of turbulence and thus requires approximate parametrization. Korolev⁵ showed analytically using a one-dimensional model that curvature and solute effects broaden droplet size spectra during condensation.

Atmospheric turbulence affects key cloud microphysical processes like condensation, evaporation, activation, and deactivation at the microscale.⁶ When cloud droplets and aerosol particles are suspended in a turbulent environment, the mass and energy transfer between the moist air and the dispersed phase (particles) becomes important. Grabowski *et al.*⁷ studied the activation process in turbulent and non-turbulent cloud models and reported that all aerosol particles activate into cloud droplets in a non-turbulent setting but only a fraction of aerosol particles do so in a turbulent environment. On the other

hand, Shawon *et al.*,⁸ who studied the dependence of aerosol-cloud interactions on turbulence in a laboratory setting, observed that turbulence causes more aerosol particles to be activated and that the activated fraction decreases monotonically as the aerosol number concentration increases. From a direct numerical simulation (DNS) study, Celani *et al.*⁹ examined the role of turbulence in the growth process of cloud droplets, and reported turbulence-induced spectral broadening. They introduced the concept of the equivalent cloud condensation nuclei (CCN) as a way to signify the importance of turbulence in the process of cloud growth. Ditas *et al.*¹⁰ measured the fluctuations in environmental supersaturation in a turbulent stratocumulus cloud layer and found that the fluctuations decrease with increasing liquid water content and droplet number concentration. The authors also estimated the fluctuations in the temperature and vertical velocity, as well as the number fraction of activated aerosol particles. Thomas *et al.*¹¹ used DNS to investigate the condensational growth and activation of aerosol particles in response to fluctuations in the environmental supersaturation resulting from turbulence. They found that aerosol particles activate in small amounts until a maximum activation is reached and larger cloud droplets co-exist with smaller unactivated aerosol particles, also known as haze particles. An interesting investigation by Prabhakaran *et al.*¹² identified three regimes of aerosol activation from their experimental and theoretical study: mean-dominated, fluctuation-influenced, and fluctuation-dominated. The authors found that turbulence tends to deactivate cloud droplets and suppress aerosol activation in the mean-dominated regime. Only one-dimensional spatial models were used in this study.

Despite recent progress, as summarized in the foregoing, significant knowledge gaps exist in our understanding of the activation and deactivation processes in turbulent clouds. For example, the effects of the initial aerosol size distributions on activation and deactivation in a turbulent cloud environment have received little attention. Although the role of turbulent fluctuations on activation has been reported,^{7-9,11,12} a systematic study of the impact of a range of turbulence levels on activation and deactivation is required. Moreover, previous investigations have limitations, including one-dimensional analysis, approximate analyses, or the limitations associated with the use of the large-eddy simulation approach. The objective of this study is to investigate these aspects of activation and deactivation with a particle-based DNS model. The study involves solving a temporal and three-dimensional spatial model of the continuum phase by use of the DNS approach, wherein all the scales of flow are resolved. The turbulence in our model is assumed to be homogeneous and isotropic,¹³ which allows us to use spatial averaging for the statistics. The Lagrangian approach is used to track the motion and growth of particles. The DNS model developed by Gao *et al.*¹⁴ is extended in the present work to incorporate the activation/deactivation processes as described by the Kappa-Kohler theory.¹⁵ We vary dry aerosol size distributions from monodisperse to polydisperse and generate different levels of turbulent kinetic energy (TKE) dissipation rate. The temporal growth and properties of aerosol particles and cloud droplets, along with the evolution of the thermodynamic fields of temperature and water vapor mixing ratio, are investigated.

In the Sec. II, we present the governing equations for the dynamics and thermodynamics of the problem, as well as the models employed to describe the cloud droplets, including the curvature and solute effects. We then summarize the numerical procedure employed

in this study and list the relevant parameters. This is followed by the presentation of the results, with concluding remarks at the end of the paper.

II. GOVERNING EQUATIONS

The equations solved are described in this section. They include the Navier–Stokes equations, which express the conservation of momentum in the continuous phase (air). We also solve the transport equations for the thermodynamic variables and the equations expressing the conservation of momentum for aerosols and clouds.

A. Turbulent velocity field

The turbulent air in the atmosphere is assumed to be governed by the physical-space Navier–Stokes equations with the Boussinesq approximation¹⁶

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ is the instantaneous fluid velocity vector, and p is the instantaneous fluid pressure. The terms in Eq. (1) are, in order, the time rate of change of velocity at a fixed point, the convective transport of momentum, the pressure force, the molecular dissipation of momentum, the buoyancy force, and the external force that is applied to sustain turbulence. Equation (2) is a statement of mass conservation for an incompressible flow, for which the density is constant with respect to space and time. The body force term $\mathbf{F}_b$ is present at all points in the model (body) and can be expressed as¹⁶

$$\mathbf{F}_b = -\mathbf{g} \left[\frac{T - T_o}{T_o} + 0.608(q_v - q_{vo}) - q_c \right], \quad (3)$$

where T is the instantaneous temperature, T_o is the domain average of T , q_v , and q_c are the instantaneous water vapor mixing ratio and liquid water mixing ratio, respectively, and q_{vo} is the domain average of q_v .

Although the gravitational body force term $\mathbf{F}_b$ could in principle maintain turbulence in the flow, the elevation (0.512 m the direction of gravity) in our microphysics model is too small to permit significant effects of gravity. Thus, $\mathbf{F}_b$ is extremely weak to sustain turbulence. Without turbulence production from shear, Dirichlet boundaries, or by baroclinic means, no natural mechanism is available in our model to prevent a complete decay of the imposed initial turbulence field. Thus, we introduce the external forcing term $\mathbf{F}_c$ in Eq. (1). The force is formulated in the Fourier space as¹⁷

$$\hat{\mathbf{f}}_c(\mathbf{k}, t) = \epsilon_{in} \sum_{\mathbf{k}_f \in \mathbf{k}} \frac{\hat{\mathbf{u}}(\mathbf{k}, t)}{|\hat{\mathbf{u}}(\mathbf{k}_f, t)|^2} \delta_{\mathbf{k}, \mathbf{k}_f}, \quad (4)$$

where ϵ_{in} is an input TKE dissipation rate, $\mathbf{k}$ is the wavenumber vector, $\hat{\mathbf{u}}(\mathbf{k}, t)$ is the Fourier-transformed velocity field, $\mathbf{k}_f$ is chosen from a wavenumber shell, and $\delta_{\mathbf{k}, \mathbf{k}_f}$ is a delta function. $\hat{\mathbf{f}}_c$ is inverse transformed into the physical space as $\mathbf{F}_c$ and used in our Navier–Stokes solver. Consistent with the Kolmogorov cascade theory of turbulence,¹³ the low wavenumber end of the turbulence energy spectrum is forced.¹⁸ The idea is that while the large scales of flow will be affected to some extent by forcing, the dynamics of the small scales are invariant, which emphasizes the need to localize the forcing within the low wavenumber end of the energy spectra. One could in principle vary the values of ϵ_{in} for the purpose of generating different turbulence intensities (TKE dissipation rates). However, preliminary work from the present study has shown a weak effect of ϵ_{in} when we transform the $\hat{\mathbf{f}}_c$ to the physical space, making it difficult to generate turbulence intensities that are significantly different. We have therefore chosen to use different bands of wavenumber forcing to generate the four levels of turbulence intensities (dissipation rates) that we investigate in this paper. We refer to these as “very high,” “high,” “medium,” and “low” levels of turbulence. The abbreviations “V,” “H,” “M,” and “L” are used respectively for these turbulence levels. The statistics for the four turbulence levels are provided in Table I. More details on the turbulence forcing are provided in Appendix A.

B. Thermodynamic variables

The thermodynamic variables of temperature, water vapor mixing ratio, and supersaturation need to be calculated alongside the flow velocity. The equations for temperature and water vapor mixing ratio are written in the Eulerian framework as¹⁴

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T, \quad (5)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_v \nabla^2 q_v, \quad (6)$$

where C_d is the rate of mass exchange between liquid water and water vapor and is defined below in Eq. (16). The terms in Eq. (5) are, in order, the time rate of change of temperature at a fixed point, the convective transport of heat, the latent heat, and the molecular diffusion of heat. Equation (6) can be explained similarly. The equation set in Eqs. (5) and (6) should be contrasted with those in one of the earlier studies of this problem.¹⁹ In the latter, the base solutions correspond to the constant density, hydrostatic conditions in the atmosphere. The fluctuating fields were generated by perturbing the base conditions. The combined field (mean plus fluctuating) is solved in our approach. Thus, whereas the mean thermodynamic fields were contained in the base solutions in Ref. 19, they are combined with the fluctuating thermodynamic fields in our approach. Moreover, in our model, the mean

TABLE I. Statistics of the turbulent fields at 6 s.

Turbulence level	$\bar{h}$ (m ² /s ²)	ϵ (m ² /s ³)	u_{rms} (m/s)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
Very high	0.0165	0.011 72	0.1817	0.0252	0.000 73	0.181	534.950	304.988	2910.96
High	0.007 68	0.002 93	0.1239	0.0343	0.001 03	0.229	336.939	283.698	1890.67
Medium	0.003 97	0.000 72	0.0890	0.0497	0.001 47	0.347	211.092	295.192	2061.35
Low	0.001 83	0.000 19	0.0605	0.0667	0.002 00	0.423	134.200	269.107	1706.74

values of the thermodynamic variables come directly from the initial conditions. The convective terms in Eqs. (5) and (6) are of leading order and are of the same order as the local temporal change terms. In transport phenomena, the convective and temporal evolution terms do not create or destroy momentum or species such as T or q_v . They only move the latter from one place to another. Hence, the T and q_v are primarily passive tracers in our problem. Since we apply triply periodic boundary conditions in our studies (Sec. III A), there is no boundary layer behavior. Otherwise, the small magnitude of the diffusion terms would lead to a (mathematically) stiff boundary layer behavior, making the specified initial value give way to the effects caused by a Dirichlet value at the wall, if one were specified. The absence of a no-slip condition coupled with the small magnitudes of the diffusion terms in our problem explains why the initial conditions for the thermodynamic variables serve as the mean values.

C. Description of aerosol particles and cloud droplets

The motion of each aerosol particle or cloud droplet is represented by the Lagrangian description, which consists of the kinematics of their motion and Newton's law. The simplified equations used can be written as²⁰

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad (7)$$

$$\frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (8)$$

where the subscript “ i ” denotes the i -th aerosol particle or cloud droplet, $\mathbf{V}$ is its velocity vector, and $\mathbf{X}$ is its position vector. The terms in Eq. (8) denote, in order, the time rate of change of velocity, the Stokes drag force, and the gravitational force. The particle response time τ_p is given by Ref. 20

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}. \quad (9)$$

Equation (9) is valid for Stokes particles for which the Reynolds number based on the relative velocity between particle and bulk flow is significantly less than unity.²¹ $\mathbf{u}_i$ is the instantaneous fluid velocity vector at the position of particle i , and is obtained through a trilinear interpolation of the Eulerian field. Note that relative to the model in Ref. 6, the fact that $\rho_a/\rho_l \sim 10^{-3}$ is used to neglect two relative acceleration terms in Eq. (8). The Basset history term is also neglected on the basis of a small relative acceleration. In Ref. 22 and in most DNS studies in cloud physics, the change in cloud droplet radius as a result of condensation (or evaporation) is described by Ref. 22

$$\frac{dr_i(t)}{dt} = \frac{K_i}{r_i(t)} (S_e(\mathbf{X}, t))_i, \quad (10)$$

where S_e is the environmental supersaturation, which is related to q_v and T by

$$S_e = \frac{q_v}{q_{v,s}} - 1, \quad (11)$$

where $q_{v,s}$ is the saturation water vapor mixing ratio, which can be calculated from Ref. 23

$$q_{v,s} = 621.97 \frac{p_{sat}}{p - p_{sat}}. \quad (12)$$

p_{sat} is the saturation water vapor pressure, which is obtained from Ref. 24

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right). \quad (13)$$

The S_e is an Eulerian quantity in Eq. (11) but interpolated to the particle position in Eq. (10). Also, K is a growth parameter that depends on temperature and pressure and is expressed as¹⁴

$$K = \frac{1}{\left(\frac{L_h}{R_v T} - 1\right) \frac{L_h \rho_l}{k_T T} + \frac{\rho_l R_v T}{\mu_v p_{sat}}}. \quad (14)$$

The temporal rate of change in the mass of the particle population during condensation (or evaporation) can be written as¹⁴

$$\frac{d(m_l(\mathbf{x}, t))}{dt} = 4\pi\rho_l \sum_{i=1}^n r_i^2(t) \frac{dr_i(t)}{dt} = 4\pi\rho_l \sum_{i=1}^n K_i r_i(t) (S_e(\mathbf{X}, t))_i, \quad (15)$$

where m_l is the mass of liquid water per grid cell and n is the number of cloud droplets inside each grid cell. C_d can be determined as¹⁴

$$C_d(\mathbf{x}, t) \equiv \frac{d(m_l(\mathbf{x}, t))}{m_a dt} = \frac{4\pi\rho_l}{\rho_a a^3} \sum_{i=1}^n K_i r_i(t) (S_e(\mathbf{X}, t))_i, \quad (16)$$

where m_a is the mass of air in each grid cell that has a volume of a^3 . q_c , which appears in Eq. (3), is modeled as¹⁴

$$q_c(\mathbf{x}, t) = \frac{4\pi\rho_l}{3\rho_a a^3} \sum_{i=1}^n r_i^3(t). \quad (17)$$

D. Consideration of curvature and solute effects

Previous studies^{4,11} in cloud microphysics have considered the curvature and solute effects by applying the Kappa-Kohler theory.¹⁵ However, these studies were not DNS. Consistent with those and extending the DNS study by Gao *et al.*,¹⁴ we modify Eq. (10) to obtain Eq. (23) below to include the particle equilibrium saturation ratio s , which is described by

$$s = \frac{r^3 - r_d^3}{r^3 - r_d^3(1 - \kappa)} \exp\left(\frac{2\sigma_l M_l}{RT\rho_l r}\right). \quad (18)$$

Here, r_d is the dry radius before growth, r is the wet radius, and κ is the hygroscopicity of solute (dry aerosol). Let A be a parameter defined as¹⁵

$$A = \frac{2\sigma_l M_l}{RT\rho_l}. \quad (19)$$

By taking the natural logarithm of both sides of Eq. (18), we get

$$\ln(s) = \ln\left(\frac{r^3 - r_d^3}{r^3 - r_d^3(1 - \kappa)}\right) + \frac{A}{r}.$$

From the Taylor expansion

$$\ln(x) \equiv (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots, \quad 0 < x < 2,$$

and ignoring the higher order terms, we have the simple approximation

$$\ln(x) \approx (x-1), \quad \text{for } 0 < x < 2,$$

where x can be s or $(r^3 - r_d^3)/(r^3 - r_d^3(1 - \kappa))$, and the convergence restriction on the expansion is met for the problem being analyzed in this paper. Using this approximation and simplifying, we get

$$s = 1 + \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1 - \kappa)}. \quad (20)$$

The particle equilibrium supersaturation S_k can then be found as²⁵

$$S_k = s - 1 = \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1 - \kappa)}. \quad (21)$$

The first and second terms on the right-hand side (RHS) of Eq. (21) quantify the effects of curvature and solute, respectively. Note that the effects of the two terms are combined in this study. A plot of S_k vs r using Eq. (21) gives the Kohler curve.²⁵ The critical equilibrium supersaturation (S_c) and the critical radius (r_c)²⁶ can be found as the stationary point of Eq. (21), using standard gradient optimization and the fact that $r^3 \gg r_d^3$

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}, \quad S_c = \frac{2}{\sqrt{\kappa}} \left(\frac{A}{3r_d} \right)^{\frac{3}{2}}. \quad (22)$$

When curvature and solute effects are considered, the rate of change of particle radius during condensation (or evaporation) can be written as²⁷

$$\frac{dr_i(t)}{dt} = \frac{G_i}{r_i(t)} (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (23)$$

The modified growth parameter G is defined as²⁷

$$G = \frac{1}{\frac{L_h \rho_l}{k'_T T} \left(\frac{L_h}{R_v T} - 1 \right) (1 + S_k) + \frac{\rho_l R_v T}{\mu'_v p_{sat}}}, \quad (24)$$

where k'_T and μ'_v are modified thermal conductivity and modified water vapor diffusivity in air, respectively. They are modeled as²⁷

$$k'_T = \frac{k_T}{1 + \frac{k_T}{\alpha_T r \rho_a c_p} \sqrt{\frac{2\pi M_a}{RT}}}, \quad \mu'_v = \frac{\mu_v}{1 + \frac{\mu_v}{\alpha_r} \sqrt{\frac{2\pi M_l}{RT}}}. \quad (25)$$

The parameters α_T and α_c are the thermal accommodation and mass accommodation coefficients, respectively. Their values are taken as unity in this work following Ref. 3. With the foregoing, the rate of change of mass during condensation (or evaporation) can be written as

$$\begin{aligned} \frac{d(m_l(\mathbf{x}, t))}{dt} &= 4\pi \rho_l \sum_{i=1}^n r_i^2(t) \frac{dr_i(t)}{dt} \\ &= 4\pi \rho_l \sum_{i=1}^n G_i r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i, \end{aligned} \quad (26)$$

where the condensation (evaporation) rate C_d is

$$C_d(\mathbf{x}, t) = \frac{d(m_l(\mathbf{x}, t))}{m_a dt} = \frac{4\pi \rho_l}{\rho_a a^3} \sum_{i=1}^n G_i r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (27)$$

The critical radius criterion [Eq. (22)] is used to determine activation or deactivation. For an aerosol particle to be activated into a cloud droplet, it needs to grow beyond the critical radius. Similarly, for a cloud droplet to be deactivated into an aerosol particle, it has to shrink below the critical radius. According to Eq. (23), particle growth is positive when $S_e > S_k$ and negative if $S_e < S_k$. So, an aerosol particle, for which $r_d < r \leq r_c$, will grow as long as $S_e - S_k > 0$. It will activate into a cloud droplet when r becomes larger than r_c . Similarly, a cloud droplet, for which $r > r_c$, will shrink as long as $S_e - S_k < 0$, and deactivate into an aerosol particle when r becomes smaller than r_c .

E. Effect of aerosol size distributions

To explore potential effects of initial aerosol size distributions, both monodisperse and polydisperse dry aerosol size distributions are investigated in this work. For the monodisperse r_d distributions, all the dry aerosol particles have the same size (0.1 μm). For the polydisperse distributions, we assume lognormal, which is defined as²⁸

$$f(r_d) = \frac{1}{r_d \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln r_d - \mu)^2}{2\sigma^2}\right), \quad (28)$$

where $f(r_d)$ is the normalized size distribution function and μ and σ are the mean and standard deviation of r_d , respectively.

III. SIMULATION DETAILS

We conduct studies for two scenarios: scenario I and scenario II. The first is focused on condensation and activation of aerosol particles into cloud droplets, while the second is primarily on evaporation and deactivation of cloud droplets into aerosol particles. Note that activation and deactivation may occur simultaneously in the natural clouds. Given that our model's domain size is much smaller relative to a cloud, it will be difficult to observe activation and deactivation at the same time and location. Hence, we design two separate scenarios that correspond to two different initial conditions. In the first scenario, the cloud volume mixes with saturated ambient air and activation prevails. In the second scenario, the cloud volume mixes with dry (subsaturated) air and deactivation dominates.

A. The numerical model

The statistics of the turbulent velocity field at 6 s are listed in Table I to show the magnitudes of the turbulence statistics that are involved in this study. These statistics include the turbulent kinetic energy (TKE), $\bar{\epsilon}$, its dissipation rate, ϵ , the root mean square velocity fluctuation, u_{rms} , the Kolmogorov length scale $\eta = (\nu^3/\epsilon)^{1/4}$, the Taylor micro-scale, $\lambda = \sqrt{10(\nu)(\bar{\epsilon})/\epsilon}$, the length scale characterizing large eddies, $l_0 = (\bar{\epsilon})^{3/2}/\epsilon$, the forcing Reynolds number, $Re_f = (\epsilon^{1/3})(k_{min}^{-4/3})/\nu$, where $k_{min} \equiv 2\pi/L = 12.27$ is the minimum wavenumber, the Taylor-scale Reynolds number, $Re_\lambda = (u_{rms})(\lambda)/\nu$, and the Reynolds number based on l_0 , $Re_{l_0} = (u_{rms})(l_0)/\nu$. Table I shows that the TKE varies between 0.001 83 and 0.0165 m^2/s^2 , with corresponding dissipation rate values of 0.000 19 and 0.011 72 m^2/s^3

and u_{rms} values of 0.0605 and 0.1817 m/s. The ranges for Taylor microscale λ , integral scale l_0 , and their respective Reynolds numbers are (0.0252, 0.0667) m, (0.181, 0.423) m, (134.200, 534.950), and (1707, 2911), respectively. The ranges for forcing Reynolds number Re_f and Kolmogorov length scale η are (134.20, 534.95) and (0.000 73, 0.0020) m, respectively. These values are within the scope of the ambient microscale cloud environment.³⁰

The physical model simulated in this work is that of a cubic domain (box) inside which turbulence, cloud, and aerosol interact. A triply periodic condition is assumed on the boundaries of the cube for the primary dependent variables. It is well known in turbulence theory that the larger the ratio of box length to the integral length scale of turbulence, the less restrictive the assumption of periodicity. With due consideration for the physical microscale problem in the atmosphere,³⁰ the computational domain is set to 0.512^3 m³ in the present study, as in Ref. 14. The domain is divided equally into cloudy and clear-air fractions (Fig. 1) at the start of the simulations. The cloudy region is assumed to always be supersaturated while the clear-air region can be

just saturated or subsaturated. Initially, only the former contains cloud or aerosol particles, but subsequent turbulent mixing causes the particles or droplets to spread to all parts of the domain. A slab-like cloud configuration (Fig. 1) is assumed for the initial field,¹⁴ to enable a sharp transition (Fig. 2) in the temperature and vapor mixing ratio at the cloud/clear-air interface.

The thermodynamic fields and particles are initialized as follows. In scenario I, the cloudy region is supersaturated, while the clear-air region is just saturated. The initial vapor mixing ratio field is specified as¹⁴

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d \\ +L \times 0.5 \times 0.5 \\ q_{v,e}, & elsewhere. \end{cases} \quad (29)$$

The value of $q_{v,s}$ in Eq. (12) is 3.872 g/kg. Above, $q_v^{max} = 1.02 \times q_{v,s} = 3.949$ g/kg, and $q_{v,e} = 3.872$ g/kg. The length, $d \equiv L/2$ is

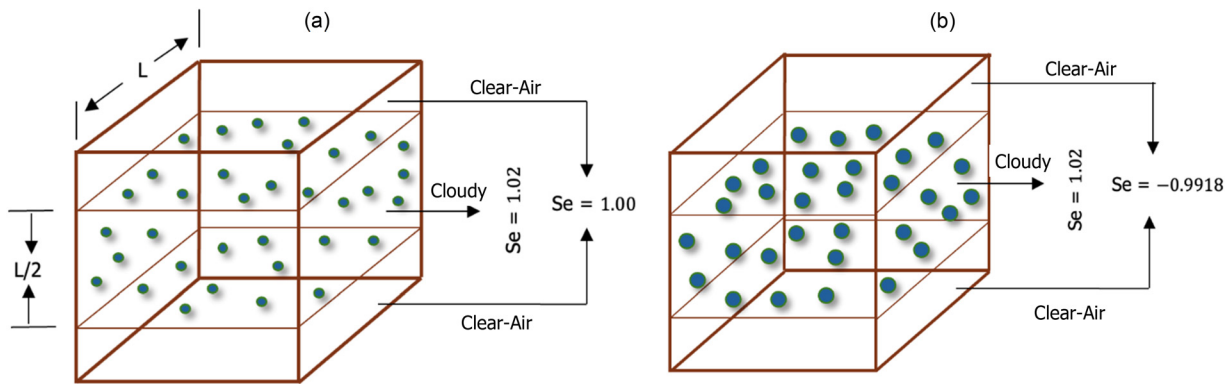


FIG. 1. Initial configurations for (a) scenario I and (b) scenario II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. The cloudy region is initially 2% supersaturated in both scenarios while the clear-air region is just saturated in the first and −99.18% subsaturated in the second. The spheres represent aerosol particles in scenario I and cloud droplets in scenario II.

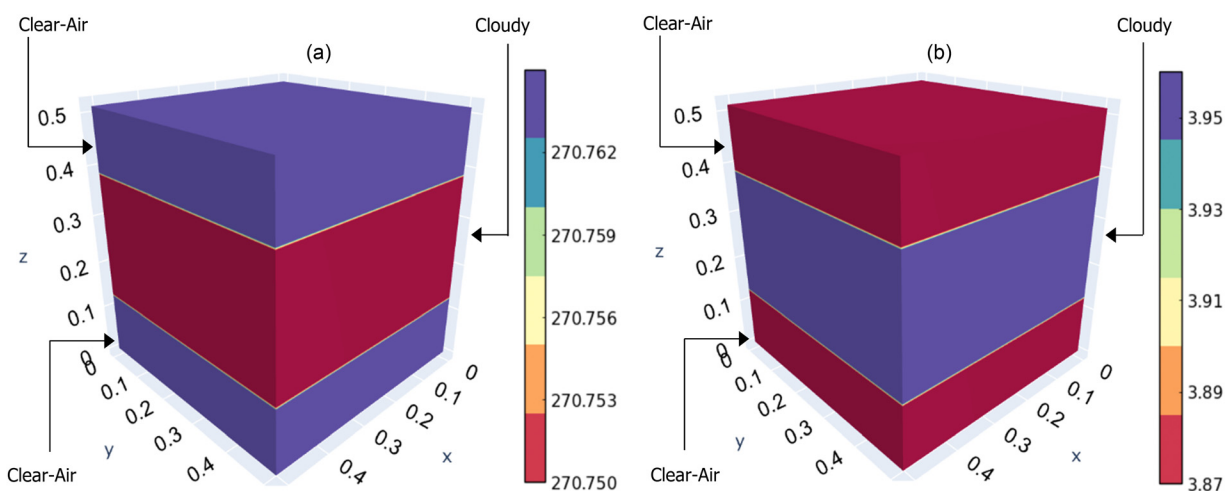


FIG. 2. Contour plots of (a) initial temperature and (b) initial vapor mixing ratio, in scenario I. A sharp transition is visible in the temperature and vapor mixing ratio profiles. The initial profiles in scenario II are similar but have different magnitudes.

TABLE II. Cases in scenario I and scenario II. Case H has “high” turbulence level and “monodisperse” dry size distribution but does not model curvature and solute effects. Case H–M is case H but with inclusion of curvature and solute effects. Cases H–L, V–L, M–L, and L–L have “lognormal” dry size distribution and include curvature and solute effects. Their turbulence levels are “high,” “very high,” “medium,” and “low,” respectively. The initial size distribution of cloud droplets is monodisperse.

Case	Dry aerosol size (μm)	TKE dissipation rate (m^2/s^3)	Turbulence level	Curvature and solute effects	Initial particle size in scenario I (μm)	Initial particle size in scenario II (μm)
H	0.1	2.93×10^{-3}	High	No	0.1	15
H–M	0.1	2.93×10^{-3}	High	Yes	0.1	15
H–L	0.1 ± 0.015	2.93×10^{-3}	High	Yes	0.1 ± 0.015	15
V–L	0.1 ± 0.015	1.17×10^{-2}	Very high	Yes	0.1 ± 0.015	15
M–L	0.1 ± 0.015	7.2×10^{-4}	Medium	Yes	0.1 ± 0.015	15
L–L	0.1 ± 0.015	1.9×10^{-4}	Low	Yes	0.1 ± 0.015	15

the width of the cloud slab. By setting $F_b = 0$ and $q_c = 0$ (in the initial field) in Eq. (3), the temperature field is initialized as¹⁴

$$T(t=0) = \langle T(t=0) \rangle - 0.608 \langle T(t=0) \rangle (q_v(z, t=0) - \langle q_v(t=0) \rangle). \tag{30}$$

At the start of the simulations, a total of 8.8×10^6 dry aerosol particles are randomly placed in the cloudy region, giving a number concentration of 65.565 cm^{-3} . The simulations start from dry aerosol radii. We simulate six cases: H, H–M, H–L, V–L, M–L, and L–L, which are summarized in Table II. The identifiers “H,” “V,” “M,” and “L,” when positioned before the dash (–) imply “high,” “very high,” “medium,” and “low” turbulence intensities, respectively. The identifiers “M” and “L,” when positioned after the dash (–) indicate “monodisperse” (M) and “lognormal” (L) dry aerosol size distributions. Dry aerosol radius (r_d) is $0.1 \mu\text{m}$, and the distributions are monodisperse for cases H and H–M. For case H–L, the dry aerosol size distribution is lognormal (L), the geometric mean (μ) is $0.1 \mu\text{m}$, the standard deviation (σ) is $0.015 \mu\text{m}$, the maximum r_d is $0.22 \mu\text{m}$, and the minimum r_d is $0.04 \mu\text{m}$. The curvature and solute effects are neglected in case H. Equation (10) is used to model the particle growth in case H while Eq. (23) is used in all other cases. Using the mean value of $0.1 \mu\text{m}$, the initial volume fraction occupied by aerosol particles is approximately equal to $2.75 \times 10^{-11}\%$.

This gives a Stokes number (S_t) of 6.37×10^{-8} in scenario I. Under this condition, the particle is essentially a tracer that is freely carried around by the continuous air phase.

In scenario II, the cloudy region is supersaturated, but the clear-air region is subsaturated. The same configuration as in Eq. (29) is used except that $q_{v,e} = 0.03138 \text{ g/kg}$, indicating a subsaturated environment. The temperature field is also initialized like in Eq. (30). A total of 8.8×10^6 cloud droplets, each having a radius of $15 \mu\text{m}$, are randomly placed in the cloudy region at the start. Dry aerosols (r_d) are assumed to be present in the core of cloud droplets (r). With r_d known, r_c can be calculated using Eq. (22). Note that r_d and r_c at each particle location do not change with time but r evolves with time. The initial volume fraction occupied by cloud droplets in scenario II is $9.27 \times 10^{-5}\%$ and the S_t is 1.43×10^{-3} . We simulate the six cases (Table II) in scenario II. Note that particles have a zero velocity at the initial time.

B. List of parameters

We tabulate the constants and initial values of the relevant parameters in Table III. A hygroscopicity parameter (κ) of 0.61 is used to represent ammonium sulfate aerosols.³¹ The mean velocity fields are zero in our model due to the assumption of HIT. The domain is

TABLE III. List of parameters and constants.

Symbol	Value	Unit	Symbol	Value	Unit
$q_{v,s}$	3.8725	g/kg	$q_{v,e}$	3.8725 (scenario I) 0.03138 (scenario II)	g/kg
q_v^{max}	3.949	g/kg			
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	0.92	kg/m ³	R	8.314	J/mol/K
ρ_l	1000	kg/m ³	p	71446	N/m ²
$\langle T(t=0) \rangle$	270.75	K	R_d	287.0	J/kg/K
ν	1.5×10^{-5}	m ² /s	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	p_{sat}	512.617	N/m ²
μ_T	2.16×10^{-5}	m ² /s	σ_l	0.072	N/m
μ_v	2.16×10^{-5}	m ² /s	M_l	0.018	kg/mol
κ	0.61	...	M_a	0.029	kg/mol
k_{min}	12.27	1/m	S_t	6.37×10^{-8} (scenario I)	...
k_{max}	1571	1/m		1.43×10^{-3} (scenario II)	...

much smaller relative to an actual cloud. All these assumptions are summarized in Appendix B. The mean variables do not vary with height in our microscale DNS model, with supporting analysis provided in Appendix C.

C. Numerical implementation

The FronTier computational fluid dynamics software package³² has been used to solve the model equations used in this study. The convective terms are solved using the finite-difference based fifth order weighted essentially non-oscillatory (WENO) scheme. A projection scheme is used, and the Poisson equation is solved for the pressure field. The PETSc package³³ is adopted to solve the systems of linear equations that arise from the discretization of the diffusion terms in Eqs. (1), (5), and (6). The Crank–Nicolson scheme is used as the time integration method. The ordinary differential equations for particle propagation, Eqs. (7) and (8), are solved with the implicit Euler method. Parallelization is achieved based on domain decomposition and the message passing interface (MPI) protocol.

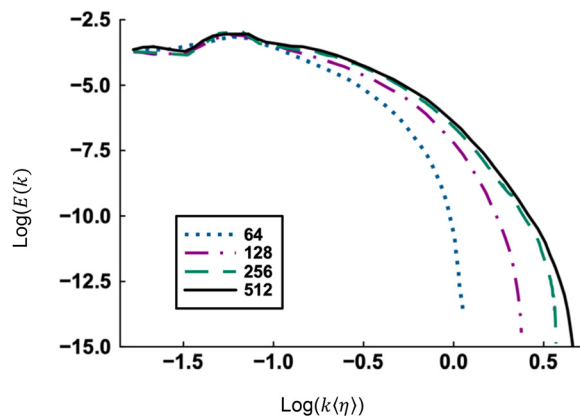


FIG. 3. The TKE spectra for four grid resolutions in logarithmic scale. The plots represent case H–L at 3 s. Note that wavenumbers (k) in the abscissa are multiples of k_{min} and are in the radial coordinate direction. The legend labels indicate the number of grid points in one coordinate direction.

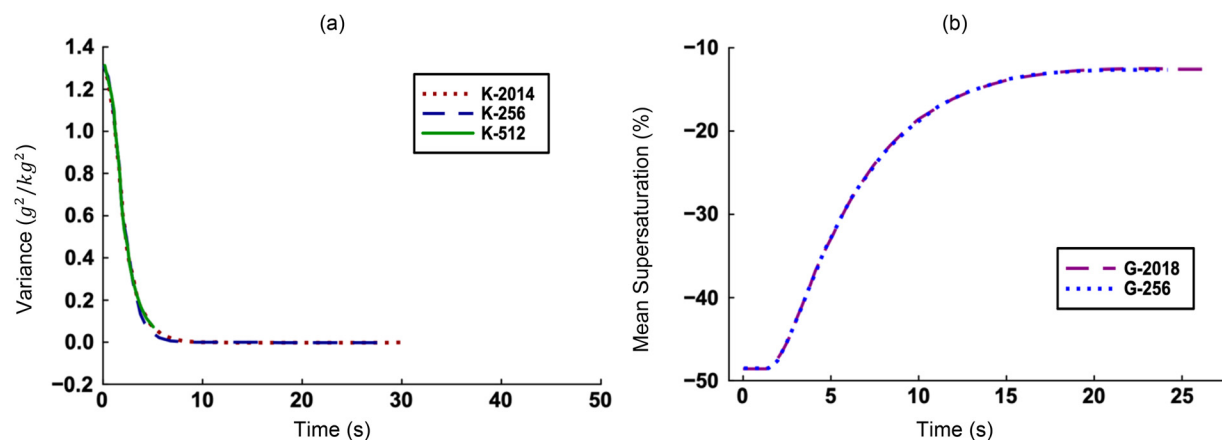


FIG. 4. Time evolution of the (a) variance of water vapor mixing ratio and (b) mean supersaturation.

The $k_{max}\eta$ value for cases H, H–M, and H–L is 1.62, where k_{max} is the maximum wavenumber and η is the Kolmogorov spatial scale. The corresponding values for cases V–L, M–L, and L–L are 1.15, 2.30, and 3.14, respectively. Based on the recommended lower bound of 1.5 for this parameter ($k_{max}\eta$),¹³ we should have good resolution of the small scales of turbulence except perhaps for case V–L. To assess the impact of grid resolution, we compared the TKE spectra, $E(k)$ ³⁴ for simulations that use 64^3 , 128^3 , 256^3 , and 512^3 grid points, for the domain size of 0.512^3 m³. Figure 3 shows that the spectra are essentially identical for 256^3 and 512^3 for the range of wavenumbers that significantly contribute to the integral of this quantity to give the TKE. Hence, we conclude that the 256^3 mesh gives an acceptable compromise between computational cost and accuracy.

As mentioned in Gao *et al.*,¹⁴ one reason for developing a finite-difference based DNS model is the need to model the sharp cloud/clear-air interface [Eq. (29)] which is often found in cumulus clouds.³⁵ We are aware that the pseudospectral method is more accurate than the finite difference method for solving the incompressible Navier–Stokes equations.³⁶ However, the pseudospectral method is highly susceptible to sharp gradients of the flow field,³⁷ where it develops unphysical oscillations (Gibbs phenomenon). The discontinuities between the cloudy and clear-air regions in our model (Fig. 2) will pose difficulties for the pseudospectral scheme. In a DNS study of clouds, Kumar *et al.*²⁰ introduced a smooth function in the initial vapor mixing ratio field to avoid numerical overshoots caused by the pseudospectral method. An example of a DNS investigation of cloud physics that used the finite difference method can be found in MacMillan *et al.*³⁸ who studied the cloud chamber, for which the no-slip condition has to be imposed at the top and bottom walls, while the pseudospectral method is used in the horizontal directions where periodic boundary conditions are appropriate. In addition, our team is working on a digital twin of the cloud chamber, for which Dirichlet boundary conditions must be used with DNS, and the pseudospectral scheme is not directly applicable. Note that use of finite difference for DNS is commonplace in the physics and mechanical engineering literature.

We utilize the high-performance computing environment offered by the Perlmutter supercomputer managed by National Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA for the simulations.

D. DNS model validation

We chose the FronTier code because of its ability to simulate two-phase flows. For example, Bo *et al.*³⁹ employed it to solve the Navier–Stokes equations and track the interface separated by two incompressible fluids having different densities. Their results matched the experimentally obtained correlations of Wu and Fateh.⁴⁰ Du *et al.*⁴¹ applied the code to calculate rotating slotted disk flows, and swirling ellipsoid and reversal flow tests. They were able to produce the correct topological bifurcations. Within the context of clouds, Gao *et al.*¹⁴ developed a DNS model with Lagrangian microphysics and used an older version of the FronTier code. To validate our DNS model, we ran the updated FronTier code using the parameters of Kumar *et al.*,²⁰ wherein the latter reference used a pseudospectral code. The profiles of the variance of vapor mixing ratio are plotted in Fig. 4(a). Case K-2014 in the graph represent the data from Kumar *et al.*²⁰ (case S3). The domain volume is 0.512^3 m^3 and the grid resolution is 512^3 . Cases K-256 and K-512 are our runs, which are based on 256^3 and 512^3 grid points, respectively, but with the same parameters

and initial conditions as K-2014. We run case K-512 for only first 5 s to save computational cost, while still observing the effect of grid resolution. Excellent agreement is evident between K-2014, K-256, and K-512 in Fig. 4(a). We also run our model using the configuration of Gao *et al.*¹⁴ and compare the time evolutions of mean supersaturation in Fig. 4(b). Case G-2018 represent the data from Gao *et al.*¹⁴ (case F3) while case G-256 is our run. Complete agreement between G-2018 and G-256 is found in Fig. 4(b).

IV. RESULTS AND DISCUSSION

The results from our simulations are presented and discussed in this section. We first present the impact of dry aerosol size and the initial distributions on activation (scenario I) and deactivation (scenario II), followed by the effects of turbulence intensity on activation (scenario I) and deactivation (scenario II). The relationship between velocity turbulence, thermodynamic and microphysical fluctuations is discussed in Sec. V. Some details on turbulence forcing are provided in an Appendix.

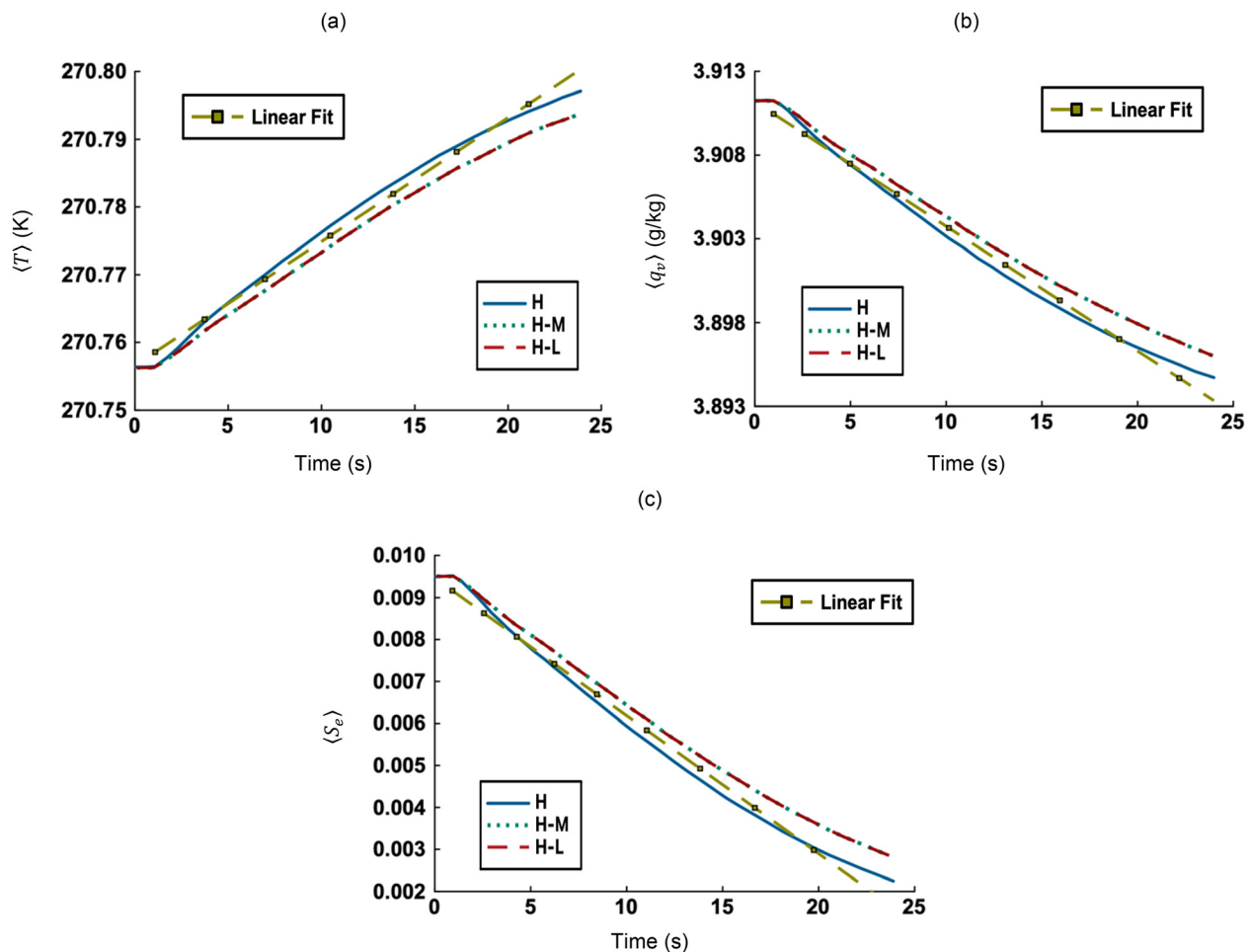


FIG. 5. Time evolution of (a) mean temperature ($\langle T \rangle$), (b) mean water vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$). The linear fit is indicated by a dashed line with “square” markers. “H” before dash (–) stands for “high” turbulence. “M” and “L” after dash (–) imply “monodisperse” and “lognormal” distributions.

A. The effects of dry aerosol size on activation and deactivation

1. Activation

Scenario I is primarily concerned with vapor condensation and aerosol activation. The temporal evolutions of the mean temperature, mean vapor mixing ratio, and mean environmental supersaturation, are plotted in Figs. 5(a)–5(c) for cases H, H-M, and H-L. These are the first three cases in Table II, which have the same turbulence intensity (high here) but differ in the treatment of the initial dry aerosol size distribution. We find in all the three cases that the mean temperature increases with time while the mean supersaturation and mean vapor mixing ratio decrease. Since water vapor condenses to become liquid water, particles grow bigger by absorbing the water while the vapor mixing ratio and supersaturation decrease. This is accompanied by an increase in the temperature due to latent heating. The attention of the reader is drawn to the small range of values on the ordinate of the graphs in Figs. 5(a) and 5(b) for the domain-averaged temperature and vapor mixing ratio. Assuming a mean value of 270.78 K, the relative variation in the temperature for case H is approximately

$(270.797 - 270.756)/270.77 \times 100\% = 0.015\%$, which is small. The corresponding value for the vapor mixing ratio is approximately 0.41%, which is also small. The change in the supersaturation over 24 s [Fig. 5(c)] is almost 108.17%, which is significant. Hence, significant variations in supersaturation are possible even when the changes in vapor mixing ratio and temperature are small. We also see that case H, which does not consider the curvature and solute effects, shows the most rapid rise (decay) in temperature (supersaturation) of all the three cases. The results for cases H-M and H-L are almost identical in Figs. 5(a)–5(c), suggesting that the initial dry aerosol size distribution does not appear to affect the evolution of the mean quantities. The mean profiles in Figs. 5(a)–5(c) follow a linear trend between 1 s to approximately 15 s. The corresponding linear regression fittings are $270.76 + 0.0018x$, $3.91 - 0.00075x$, and $0.00947 - 0.00033x$, where x is the independent variable for a particular graph. At late times, the mean thermodynamic fields change more slowly as the system has become more homogenized due to continuous mixing.

The temporal evolution of activated aerosol particle number for the same three cases are shown in Fig. 6(a). Since case H does not include curvature and solute effects, Eq. (22) cannot be applied to

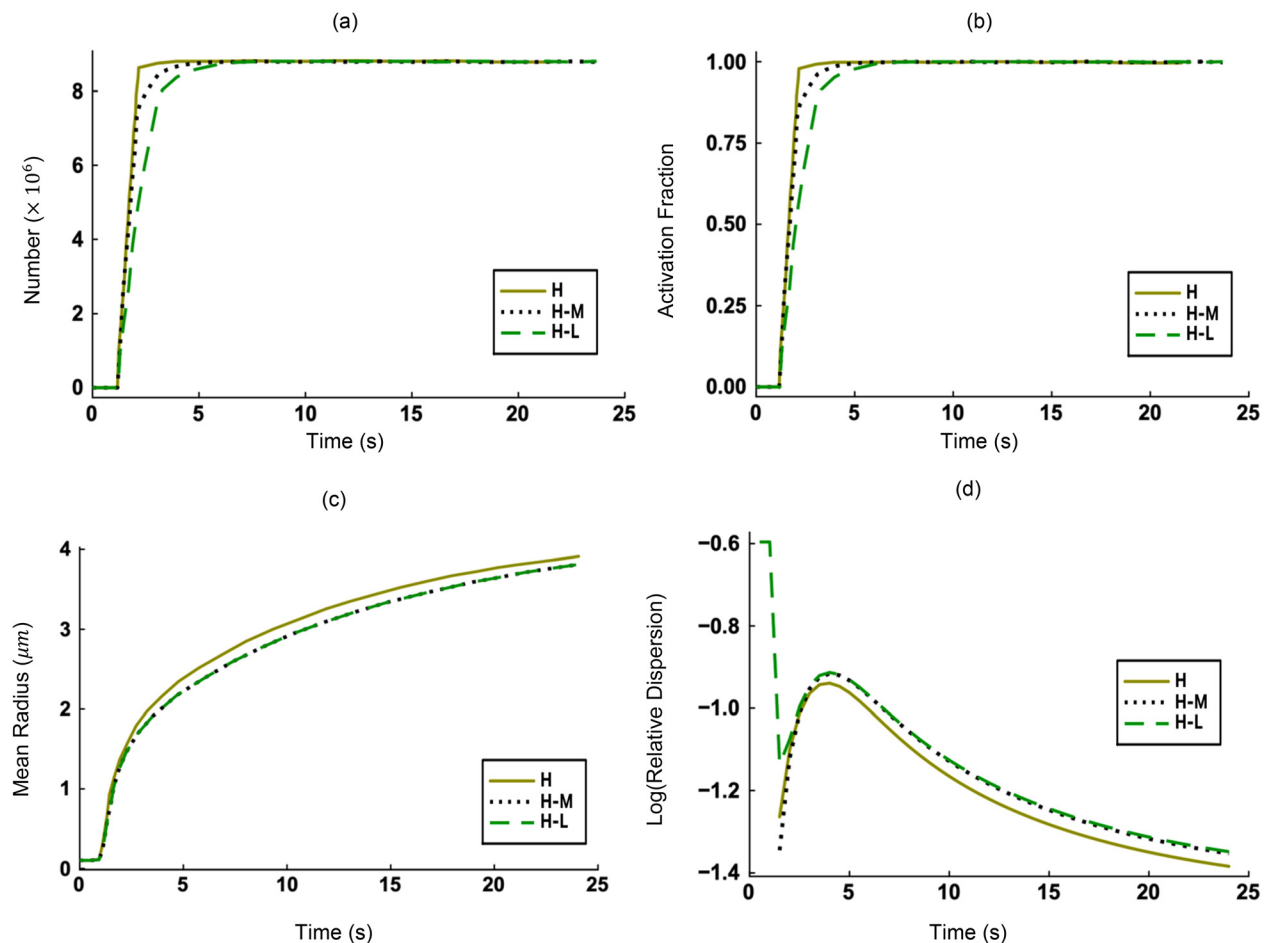


FIG. 6. Time evolution of (a) number of activated aerosol particles, (b) activation fraction, (c) mean radius of all particles, and (d) relative dispersion of particle radius in logarithmic scale in scenario I. “H” before dash (–) stands for “high” turbulence. “M” and “L” after dash (–) imply “monodisperse” and “lognormal” distributions.

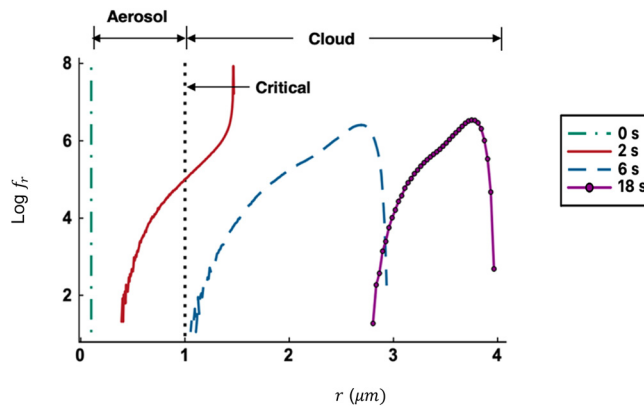


FIG. 7. Log PDFs of particle radius for case H in scenario I at selected times. The dotted line shows the critical radius (r_c).

calculate the critical radius for activation. Rather we set the critical radius to be $1 \mu\text{m}$, which is based on the empirical observation that cloud droplets are often considered to be larger than $1 \mu\text{m}$.⁴² We see in Fig. 6(a) that all aerosol particles activate into cloud droplets in approximately 4, 5, and 6 s for cases H, H-M, and H-L, respectively. The activation fraction is the ratio of activated aerosol number and total (activated and unactivated) particle number, and its time evolution is plotted in Fig. 6(b). The plot shows that activation occurs over a period of time rather than instantaneously¹¹ for monodisperse distributions (case H-M). The activation profile is steeper for case H, which has the same monodisperse distributions as H-M but no curvature and solute effects. The activation rate is slowest for case H-L on account of the lognormal distribution.⁴³ Since critical radius is directly proportional to the dry radius [Eq. (22)], larger dry aerosol particles need more time to grow beyond their critical radii and activate. Figure 6(c) shows the time evolution of the mean particle radius. We point out that “particle” is used here in a general sense and can apply to aerosol particle ($r < r_c$) or cloud droplet ($r > r_c$). It is evident that mean radius is larger for cases H compared to cases H-M and H-L. That is, the particle growth rate is slower and condensation is weaker when curvature and solute effects are included (cases H-M and H-L). The relative dispersion, defined as the ratio of standard deviation to mean, of the particle radius is plotted in Fig. 6(d). The profiles for cases H-M and H-L converge as particle radius increases, indicating that dry aerosol size distributions (monodisperse or lognormal) are less important when particles are bigger. As particles are smaller at early times, the relative dispersion of radius is higher for case H-L due to the lognormal distribution.

The probability density functions of the particle radius (f_r) at 0, 2, 6, and 18 s for case H are plotted in Fig. 7 in a log format. We see that f_r begins to broaden with time, with a tendency for convergence to a uniform, large radius at 18 s, relative to 6 s, where a wider variation in particle radii is evident as the particles grow from a uniform small (initial) radius to the larger radii at 18 s. The broadening at early times is a result of fluctuations in environmental supersaturation.⁹ Figures 8(a)–8(c) show the PDFs of radius at 2, 6, and 18 s for case H-M, which is case H with curvature and solute effects included. Aerosol is shown in solid lines in the figure while cloud is in dashed-dotted lines. Being generally of smaller size, aerosol particles are found at the lower values

of the abscissa whereas cloud droplets are at the higher values. Note the differences in the radius coordinate in Figs. 8(a)–8(c). Both aerosol particles and cloud droplets are present at 2 and 6 s but only cloud droplets can be found at 18 s. All aerosol particles activate into cloud droplets over time, and the latter continue to grow as long as the local supersaturation remains positive. The PDFs of aerosol particle radii become narrower with time and aerosols no longer exist at 18 s [Fig. 8(c)].

Figures 8(d)–8(f) show the corresponding results for case H-L. At 2 and 6 s, there is an overlap between the PDFs of aerosol and cloud radii, with the observation that the aerosol particles are sometimes larger than the cloud droplets. The distributions at 2 and 6 s should be contrasted with those for case H-M where the aerosol particles and cloud droplets do not overlap. Due to the dry aerosol particles being lognormally distributed in case H-L, there exists a distribution of critical radius instead of a single value. Thus, it is possible that a smaller aerosol particle has grown beyond its critical radius and is classified as a cloud droplet while a larger aerosol particle may not yet be large enough to cross its critical barrier and so still remains an aerosol particle. We also see that aerosols eventually vanish and only cloud droplets remain at 18 s. The PDFs of radius in Figs. 8(d)–8(f) remain skewed to the left meaning that most cloud droplets are larger than the mean radius.

2. Deactivation

Evaporation and deactivation are investigated in scenario II. The temporal evolutions of the mean temperature, vapor mixing ratio, and environmental supersaturation in Figs. 9(a)–9(c) show that, after an initial latent period, the mean temperature decreases while the mean vapor mixing ratio and the mean supersaturation increase for all the cases (H, H-M, and H-L). The temporal change is exponential until approximately 15 s, when we see a leveling-off to asymptotic values. The reduction in the mean air temperature and the increase in the mean supersaturation stem from evaporative cooling. The polynomial fits for the graphs in Figs. 8(a)–8(c) are $271.53 - 0.42x + 0.021x^2 - 0.00032x^3$, $1.80 + 0.17x - 0.008x^2 + 0.00013x^3$, and $-0.56 + 0.06x - 0.0026x^2 + 0.00003x^3$, respectively. Thus, these profiles are roughly cubic, although dominated by the linear and quadratic terms. It is important to note that for the same turbulence level, neither curvature and solute effects nor dry aerosol size distribution has any significant impact on the temporal evolution of the mean quantities. This is because, despite being negative, the magnitude of mean S_e is much larger than the value of the mean S_k (~ 0.00061) in scenario II.

Figure 10(a) shows the evolution of the number of deactivated cloud droplets for cases H, H-M, and H-L. We see that the values are identical at all times for the three cases, suggesting that the initial dry aerosol size distributions do not influence the deactivation profile. Note that while we allow the dry aerosol size distributions to be monodisperse or lognormal in both scenario I and scenario II, the initial cloud droplet size distributions are monodisperse in scenario II. The evolution of the mean radius in Fig. 10(b) shows that this quantity decreases with time for all three cases as a consequence of droplet evaporation. However, while the mean radius is indistinguishable at all times for cases H-M and H-L, its decay becomes significantly slower after approximately 7 s for case H, which is the case without curvature and solute effects. This suggests that as droplet radius r decreases and approaches r_c and r_d , curvature and solute effects begin to have an effect.

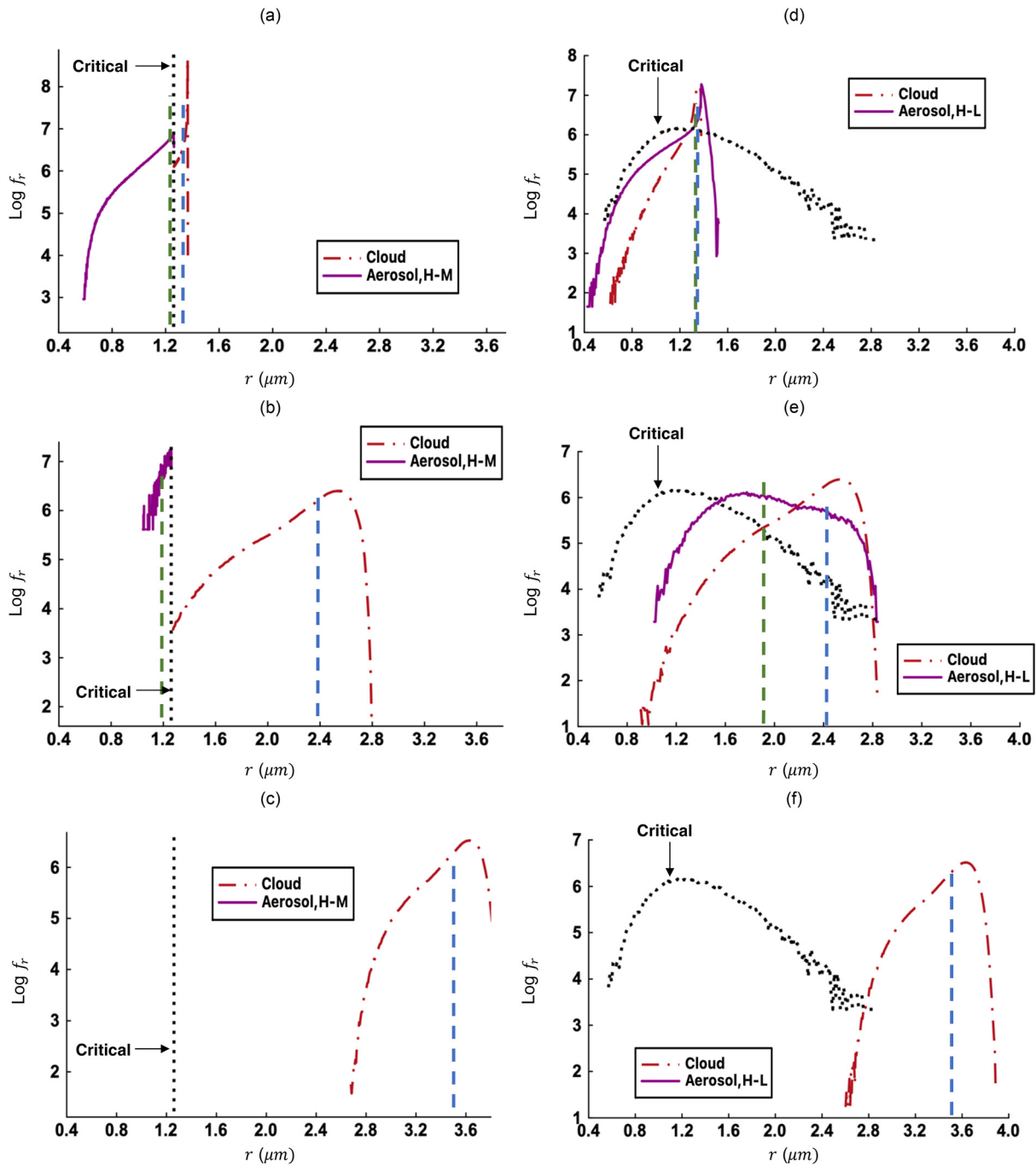


FIG. 8. Log PDFs of particle size at (a) 2 s, (b) 6 s, and (c) 18 s for case H-M in scenario I. The subplots (d), (e), and (f) are analogous for case H-L. “H” before dash (–) stands for “high” turbulence. “M” and “L” after dash (–) imply “monodisperse” and “lognormal” distributions. The dotted line shows the critical radius (r_c). The green and blue dash lines show the mean aerosol and cloud droplet radii, respectively.

The relative dispersion plots in Fig. 10(c) show different profiles for case H compared to those for the other two cases after 7 s. This confirms the importance of curvature and solute effects as the radius becomes smaller. The profiles for cases H-M and H-L are similar until

21 s. Since the r_d follows a lognormal distribution in case H-L, the relative dispersion for this case is higher when r approaches r_d (after 21 s), consistent with Fig. 6(c). The normalized droplet size distributions for case H (high turbulence but no curvature and solute effects)

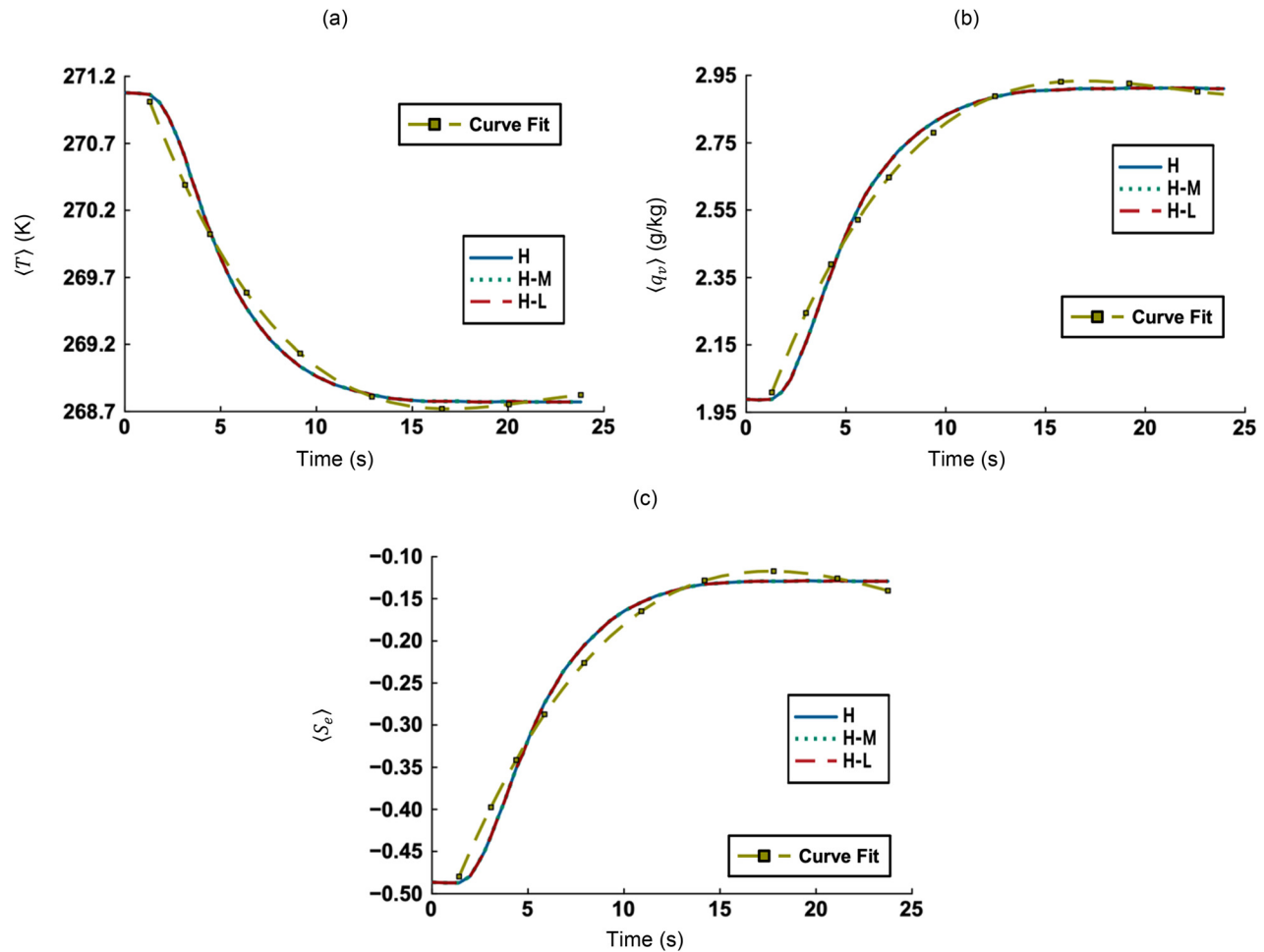


FIG. 9. Time evolution of (a) mean temperature ($\langle T \rangle$), (b) mean water vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$). The curve fit is indicated by a dashed line with "square" markers. "H" before dash (–) stands for "high" turbulence. "M" and "L" after dash (–) imply "monodisperse" and "lognormal" distributions.

are shown in Fig. 11 at 0, 4, 10, and 18 s. The cloud droplets shrink significantly with time from an initial monodisperse radius of $15 \mu\text{m}$ to values in $0 < r \leq 6.5 \mu\text{m}$ at 18 s. The PDF of radius broadens at early times, traversing the entire $0 < r \leq 15 \mu\text{m}$ range at 4 s, but subsequently becoming narrower with time as more evaporation of cloud droplets takes place. We also see that while larger radii are more probable at early times, smaller radii particles dominate at 18 s, with a mean radius of approximately $3 \mu\text{m}$.

The PDFs of particle radius for case H–M at 4, 10, and 18 s are plotted in Figs. 12(a)–12(c). This figure aims to visualize the deactivation of cloud droplets into aerosol particles. We notice that aerosol particles and cloud droplets are clearly discernible as the critical radius is single-valued due to a monodisperse dry aerosol size distribution. The corresponding results for case H–L in Figs. 12(d)–12(f) show that the f_r of aerosol particles is bimodal at the three times. That is, smaller aerosol particles ($0 < r \leq 0.5 \mu\text{m}$) form a distribution while the larger ones ($0.5 < r \leq 3 \mu\text{m}$) have another. The f_r of cloud droplets become narrower between 4 and 18 s for cases H–M and H–L. They are skewed to the left at 4 s but becomes slightly skewed to the right at 18 s for

both cases, as smaller radii cloud droplets become more probable at long time, contrasted with the converse at short time. Figure 12 shows that our model can distinguish between cloud droplets and aerosol particles at a high resolution which is not achievable by large-scale atmospheric models.

B. The effects of turbulence intensity on activation and deactivation

The results discussed so far have included only one level of turbulence: H (high turbulence intensity). In this section, we examine the activation of aerosol particles and deactivation of cloud droplets in response to variation of flow turbulence intensity.

1. Activation (scenario I)

The time evolution of the mean environmental supersaturation and its standard deviation is plotted in Figs. 13(a) and 13(b) for the four turbulence levels in scenario I. The plots in Fig. 13 show the time

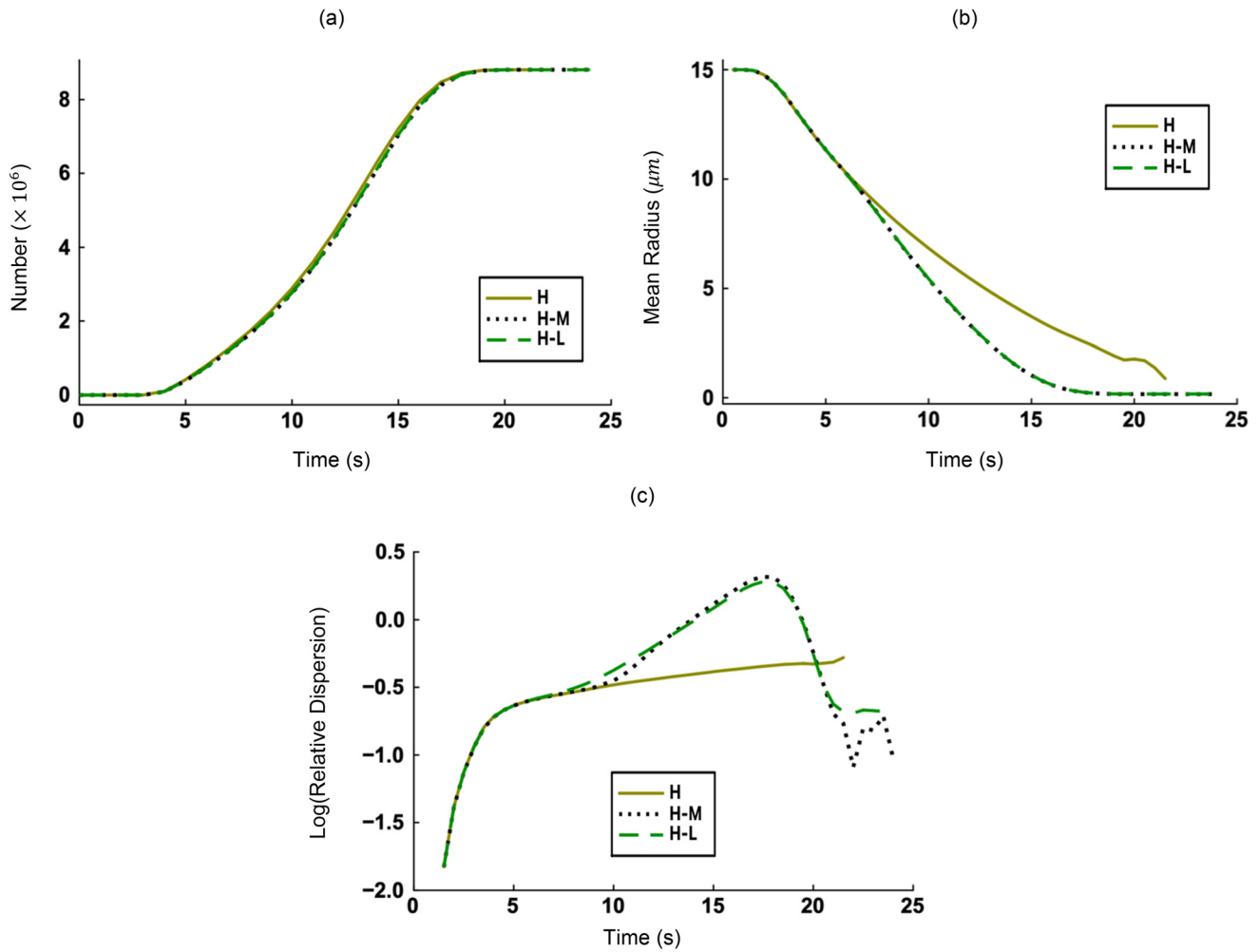


FIG. 10. Time evolution of (a) number of deactivated cloud droplets, (b) mean radius of all particles, and (c) relative dispersion of particle radius in logarithmic scale in scenario II. “H” before dash (–) stands for “high” turbulence. “M” and “L” after dash (–) imply “monodisperse” and “lognormal” distributions.

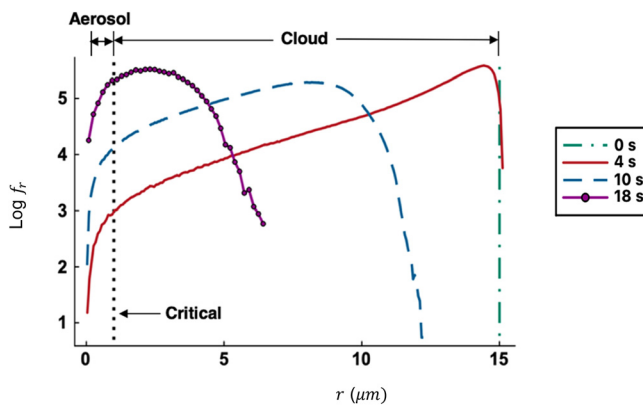


FIG. 11. Log PDFs of particle radius for case H in scenario II at selected times. The dotted line shows the critical radius (r_c).

evolution after statistical stationarity has been reached. We denote this time by τ in seconds. The transient time periods required to reach $\tau = 0$ are $t = 3$ s for case V-L, $t = 5$ s for case H-L, and $t = 6$ s for cases M-L and L-L, where t is the physical simulation time. The variance of environmental supersaturation $\sigma_{S_e}^2$ is defined as

$$\sigma_{S_e}^2 = \frac{\sum_i (S_{ei} - \langle S_e \rangle)^2}{N}, \quad (31)$$

where S_e is the instantaneous environmental supersaturation, $\langle S_e \rangle$ is the mean value, and N is the number of grid points. We see that the $\langle S_e \rangle$ values are different when the respective statistically stationary states are reached [$\tau = 0$ s in Fig. 13(a)] for the four turbulence levels, although they were the same at the start of simulations ($t = 0$ s). These differences are not caused by the early transients alone. The thermodynamic fields in Eqs. (5) and (6) are coupled with the dynamics of particles via the latent heat term C_d which accounts for particle

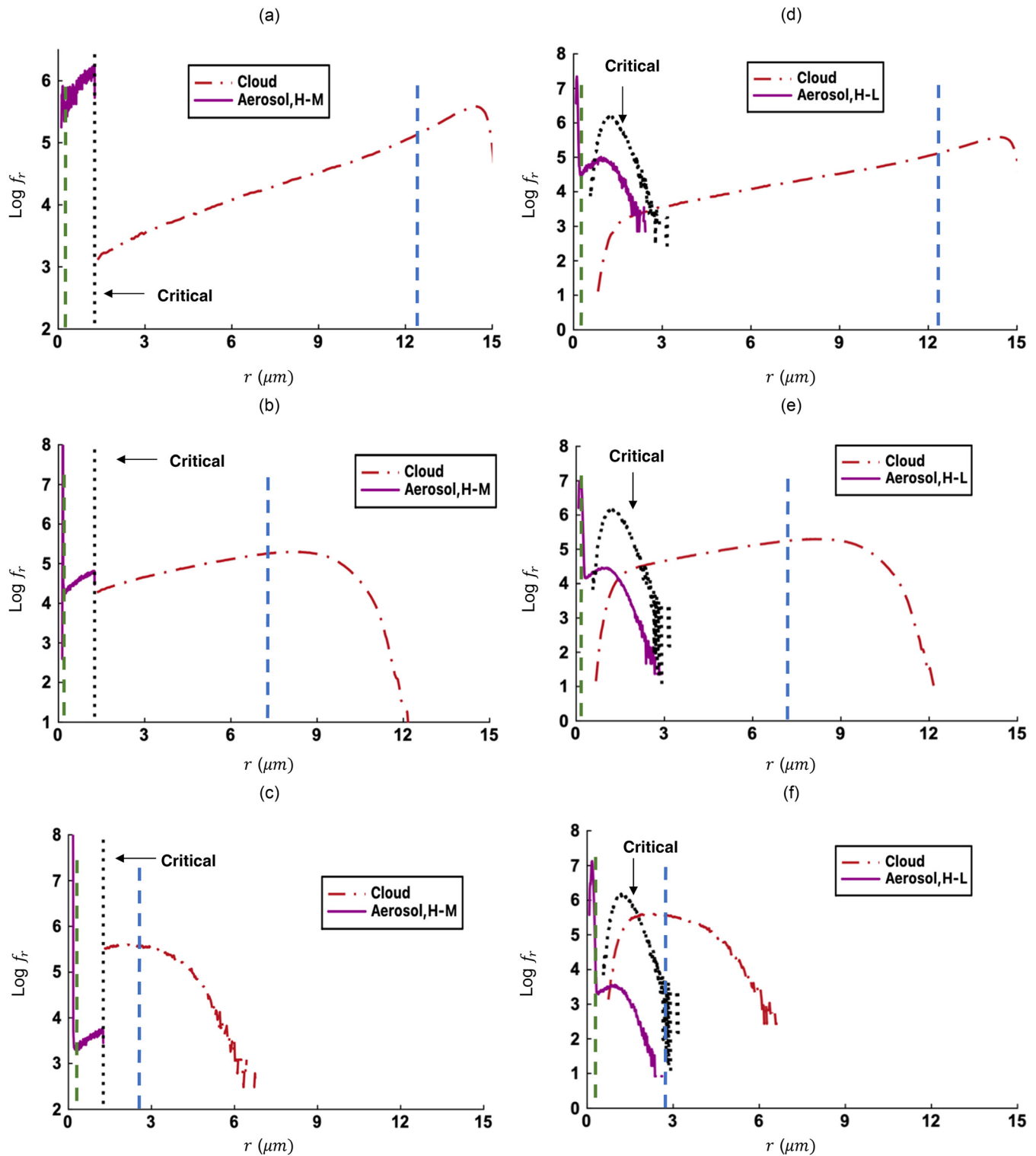


FIG. 12. Log PDFs of particle size at (a) 4 s, (b) 10 s, and (c) 18 s for case H-M in scenario II. The subplots (d), (e), and (f) are analogous for case H-L. “H” before dash (–) stands for “high” turbulence. “M” and “L” after dash (–) imply “monodisperse” and “lognormal” distributions. The dotted line shows the critical radius (r_c). The green and blue dash lines show the mean aerosol and cloud droplet radii, respectively.

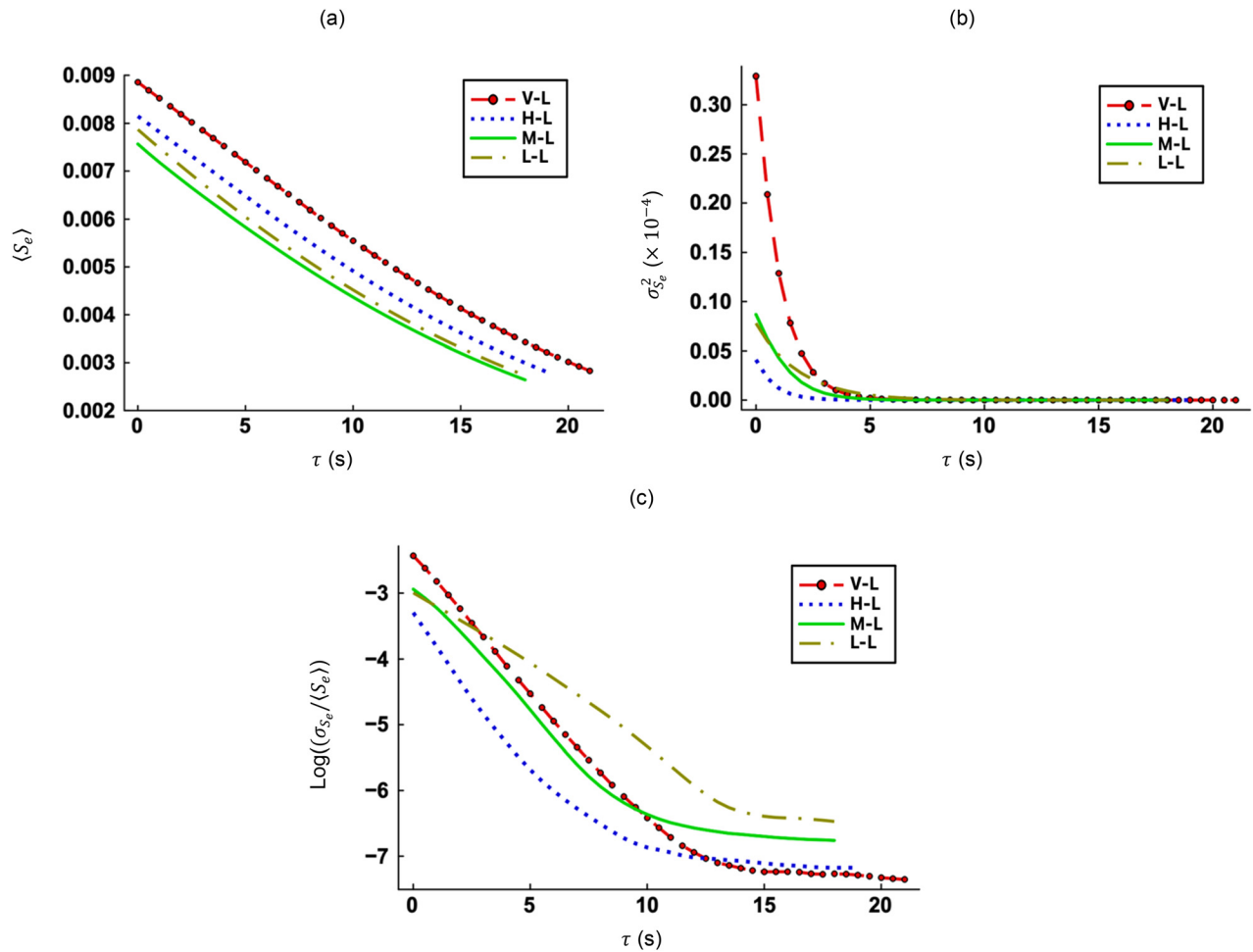


FIG. 13. Time evolution of (a) mean environmental supersaturation, (b) variance of environmental supersaturation, and (c) relative standard deviation of environmental supersaturation in a logarithmic scale in scenario I. "V," "H," "M," and "L" before dash (–) stand for "very high," "high," "medium," and "low" turbulence levels.

size [Eq. (16)]. Hence, the graphs in Fig. 13(a) demonstrate the combined effects of flow turbulence and phase change on the $\langle S_e \rangle$. We see that for the same dry aerosol size distribution (lognormal), the $\langle S_e \rangle$ is smaller for cases L–L and M–L compared to cases H–L and V–L. Also, $\sigma_{S_e}^2$ decays faster for case V–L compared to the other cases during the statistically stationary periods [Fig. 13(b)]. The results suggest that a higher level of turbulence leads to increased homogenization of the thermodynamic fields and a reduction in their fluctuations. Figure 13(c) shows that the relative standard deviation ($\sigma_{S_e}/\langle S_e \rangle$) decreases more, or the S_e values become less spread out compared to the mean, when the turbulence levels are higher (H–L and V–L).

The PDFs of S_e and S_k along with the respective mean values are plotted in Figs. 14(a) and 14(b). Aerosol particles will potentially activate when the growth is positive ($S_e > S_k$). It is evident that $\langle S_e \rangle > \langle S_k \rangle$ for case H–L at 3.5 and 6 s. So, the activation process is significantly influenced by the mean supersaturation values and has been described as mean-dominated by Prabhakaran *et al.*¹² We also see that the PDFs of S_e and S_k overlap at 3.5 s [Fig. 14(a)]. As simulations progress, fluctuations in the S_e decrease due to mixing and there is no

overlap of the PDFs at 6 s [Fig. 14(b)]. Thus, fluctuations in supersaturation have impact on activation at early times only.

Figure 15(a) shows the evolution of the number of aerosol particles that activate into cloud droplets in scenario I for the four turbulence cases. We see that this quantity is slightly lower for case V–L (very high turbulence, lognormal distribution) between 2 s and 5 s compared to the other cases. Based on this observation, turbulence appears to delay activation in our model. This is because higher turbulence leads to stronger mixing and causes most particles to grow evenly instead of some particles growing fast and crossing the activation barrier. However, all aerosol particles activate into cloud droplets eventually since activation is mean-dominated. We report in Fig. 15(b) that the mean radius of cloud droplets is smallest for case V–L throughout the simulation time. Hence, the higher the turbulence intensity the slower the droplet growth rate. The PDFs of radius in Figs. 16(a)–16(d) show that the ranges of cloud droplet radii are smallest for case V–L. We also calculate the expectation of the radius directly from the PDFs in Figs. 16(a)–16(d) based on the following definition:

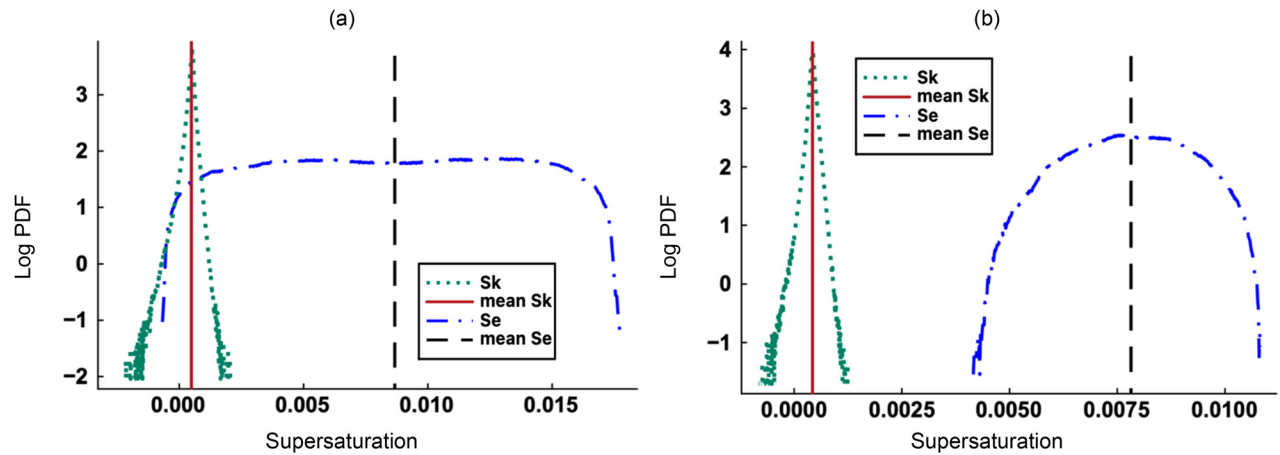


FIG. 14. Log PDFs of instantaneous supersaturation for case H-L in scenario I at (a) 3.5 s and (b) 6 s. The vertical lines indicate the mean values of the respective PDFs.

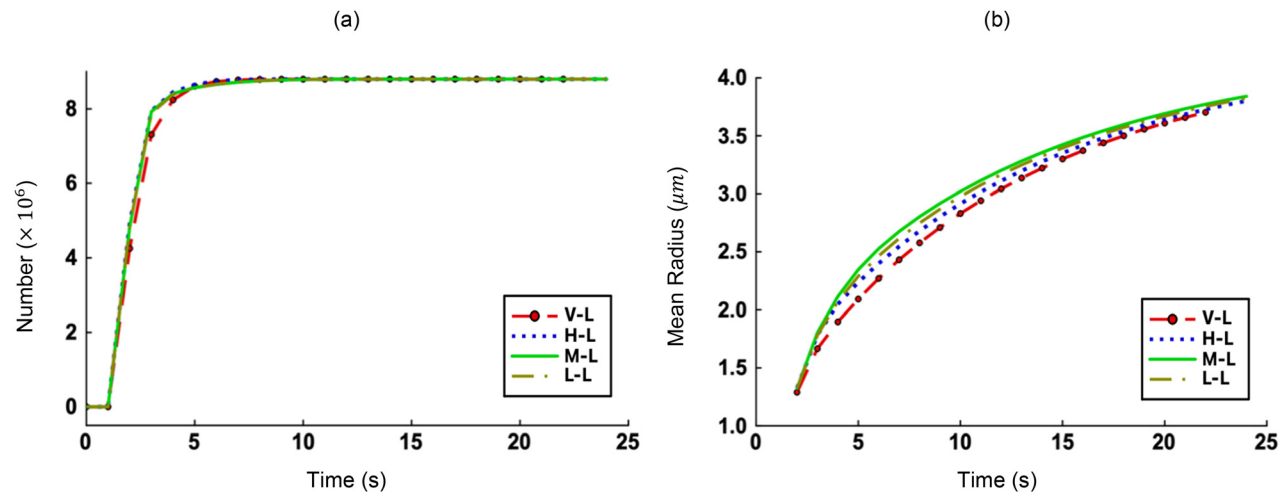


FIG. 15. Time evolution of (a) number of cloud droplets and (b) their mean radius. Subplot (b) does not include aerosol particles. "V," "H," "M," and "L" before dash (–) stand for "very high," "high," "medium," and "low" turbulence levels.

$$\langle r \rangle = \int_{\lambda=-\infty}^{\infty} r(\lambda)P(\lambda)d\lambda, \quad (32)$$

where $\langle r \rangle$ is the mean (expected) radius, r is the radius, and $P(\lambda)$ is the PDF at λ , which is an integration variable. This integral gives the mean radii in the PDFs in Figs. 16(a)–16(d) as: 2.581, 2.685, 2.809, and 2.755 μm , respectively. The corresponding mean radii at 8 s in Fig. 15(b) are 2.577, 2.681, 2.803, and 2.748 μm , respectively, which are in excellent agreement with those obtained from the PDFs.

2. Deactivation (scenario II)

Figures 17(a) and 17(b) are analogous to Figs. 13(a) and 13(b) but pertain to deactivation. We see that the mean environmental supersaturation ($\langle S_e \rangle$) increases almost cubically with time [Fig. 17(a)], in contrast to the linear decay during activation (scenario I). The $\langle S_e \rangle$ increases faster and the $\sigma_{S_e}^2$ decays faster

[Fig. 17(b)] for case V-L, which is associated with the highest turbulence level. The relative standard deviation of S_e [Fig. 17(c)] decreases faster for cases H-L and V-L compared to cases M-L and L-L. It is evident that fluctuations in supersaturation relative to the means decrease with time for all cases. Also, the decrease is faster when turbulence is stronger.

Figures 18(a) and 18(b) show the PDFs of S_e and S_k along with the respective mean values for case H-L at 5 and 10 s, respectively. Cloud droplets will potentially deactivate if the growth is negative ($S_e < S_k$). We see that $\langle S_e \rangle$ is much smaller than $\langle S_k \rangle$ in both figures. Also, most of the S_k values are closer to the $\langle S_k \rangle$. Hence, deactivation is determined mainly by $\langle S_e \rangle$ and $\langle S_k \rangle$, and is mean-dominated.¹² However, fluctuations in supersaturation can influence deactivation when the PDFs of S_e and S_k overlap [Figs. 18(a) and 18(b)]. However, this happens for a small number of particles considering that the PDF of S_k is strongly skewed to the left.

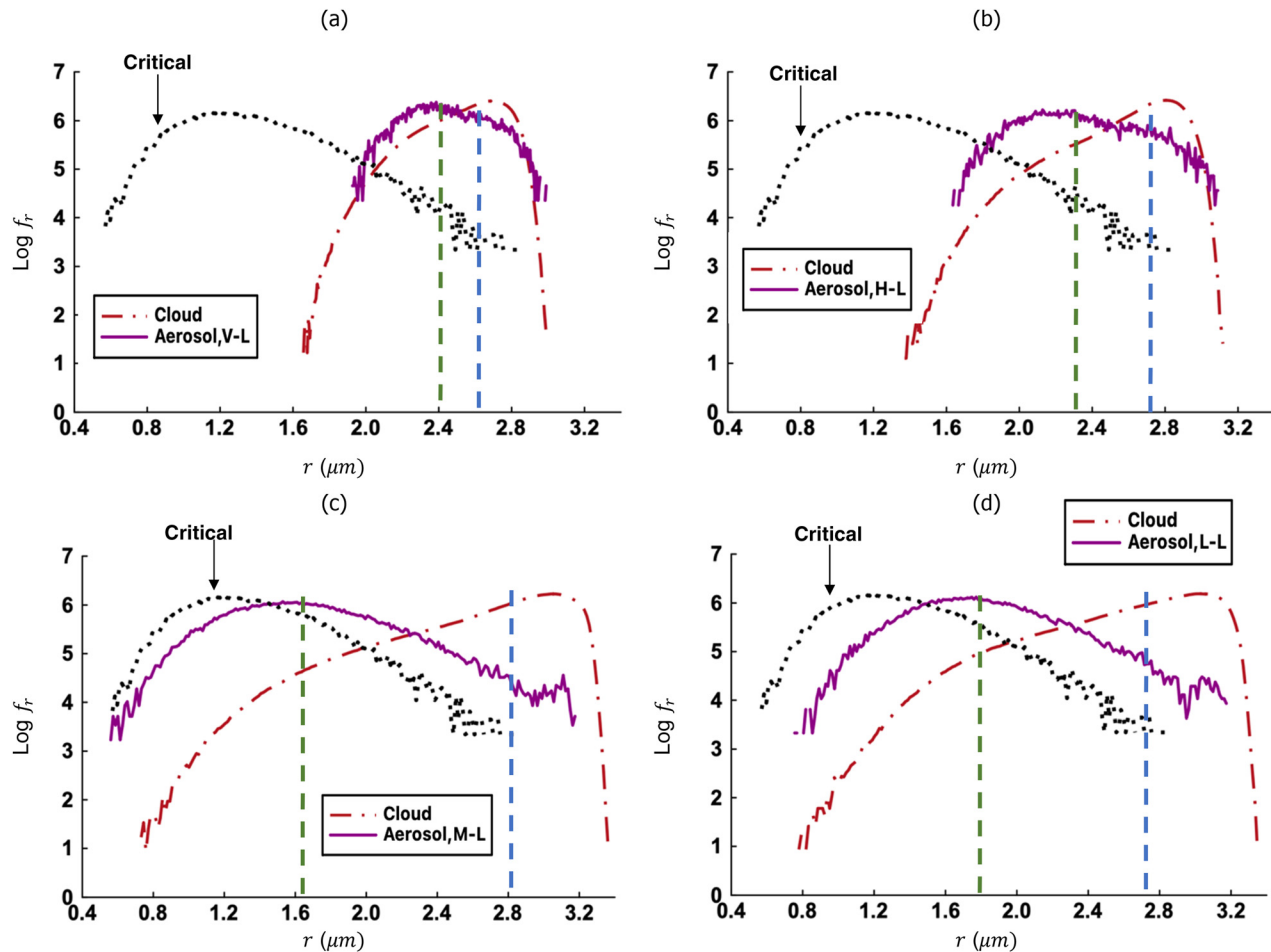


FIG. 16. Log PDF of particle size for cases (a) V-L, (b) H-L, (c) M-L, and (d) L-L, in scenario I at 8 s. “V,” “H,” “M,” and “L” before dash (–) stand for “very high,” “high,” “medium,” and “low” turbulence levels. The dotted line shows the critical radius (r_c). The green and blue dash lines show the mean aerosol and cloud droplet radii, respectively.

Figure 19(a) shows the evolution of the number of cloud droplets that shrink and deactivate into aerosol particles, while Fig. 19(b) depicts the evolution of the mean radii of deactivated cloud droplets (now aerosol particles). Based on these figures, the deactivation rate is higher, and the mean aerosol radius is larger with higher turbulence intensity. A more efficient homogenization resulting from a higher level of turbulence causes more droplets to experience partial evaporation and allows them to shrink below the critical radii. Thus, deactivation is more rapid when the turbulence intensity is high. However, we see in Fig. 19(a) that the droplet deactivation rate is marginally slower for cases V-L and H-L compared to cases M-L and L-L before approximately 8 s. Since the deactivation process starts at roughly 4 s, we attribute the behavior between 4 s and 8 s to early transients. The slopes of the deactivation profiles are comparable for cases V-L and H-L and also for cases M-L and L-L. We also see that deactivation occurs between 4 and 17 s [Fig. 19(a)] while activation happens between 1 and 6 s [Fig. 15(a)] for case H-L. The activation period is shorter than the deactivation period for the same turbulence level and same critical radii. Since initial particle size is larger in scenario II

compared to scenario I (Table II), more time is required to reach the critical radii during evaporation (deactivation).

According to Fig. 19(b), the mean radii of aerosol particles do not decrease monotonically. Theoretically, they should converge to their mean dry radii ($0.1 \mu\text{m}$ for all four cases). Figures 20(a)–20(d) show the PDFs of radius at 12 s. The maximum cloud droplet radius is largest ($14.33 \mu\text{m}$) [Fig. 20(d)] for case L-L (low turbulence, lognormal) and smallest ($9.78 \mu\text{m}$) [Fig. 20(a)] for case V-L (very high turbulence, lognormal). We calculate the mean radii of aerosol particles in Figs. 20(a)–20(d) using the PDFs. The values are, respectively: 0.192, 0.180, 0.137, and $0.129 \mu\text{m}$. The corresponding values at 12 s from Fig. 19(b) are 0.185, 0.173, 0.132, and $0.124 \mu\text{m}$. Agreement is evident, and the mean aerosol radius is larger at 12 s if the turbulence level is higher. On the contrary, the mean cloud droplet radius is smaller if turbulence is stronger as suggested by Figs. 20(a)–20(d) in which the mean droplet radii are 5.29, 5.57, 8.07, and $8.34 \mu\text{m}$, respectively. So, turbulence enhances evaporation in our model. At the same time, turbulence causes cloud droplets with radii just above the r_c to shrink to just below the r_c and be classified as aerosol particles. These particles are relatively

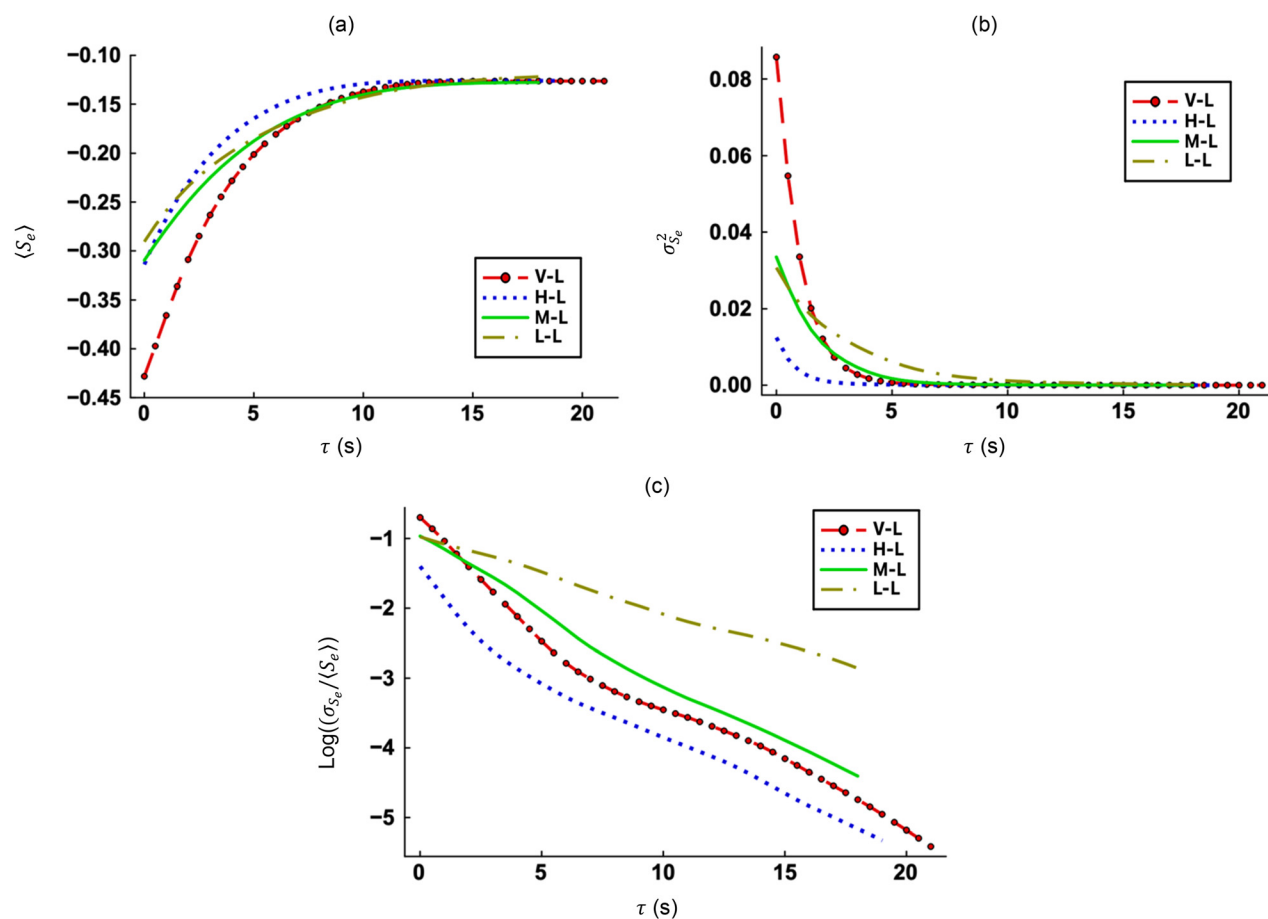


FIG. 17. Time evolution of (a) mean environmental supersaturation, (b) variance of environmental supersaturation, and (c) relative standard deviation of environmental supersaturation in a logarithmic scale in scenario II. “V,” “H,” “M,” and “L” before dash (–) stand for “very high,” “high,” “medium,” and “low” turbulence levels.

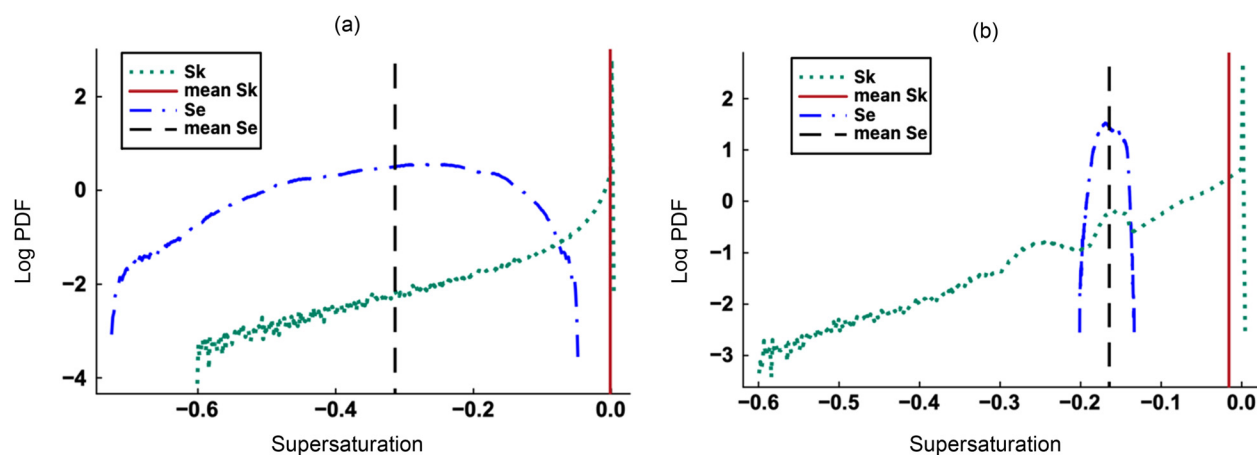


FIG. 18. Log PDFs of supersaturation for case H-L in scenario II at (a) 5 s and (b) 10 s. The vertical lines indicate the mean values of the respective PDFs.

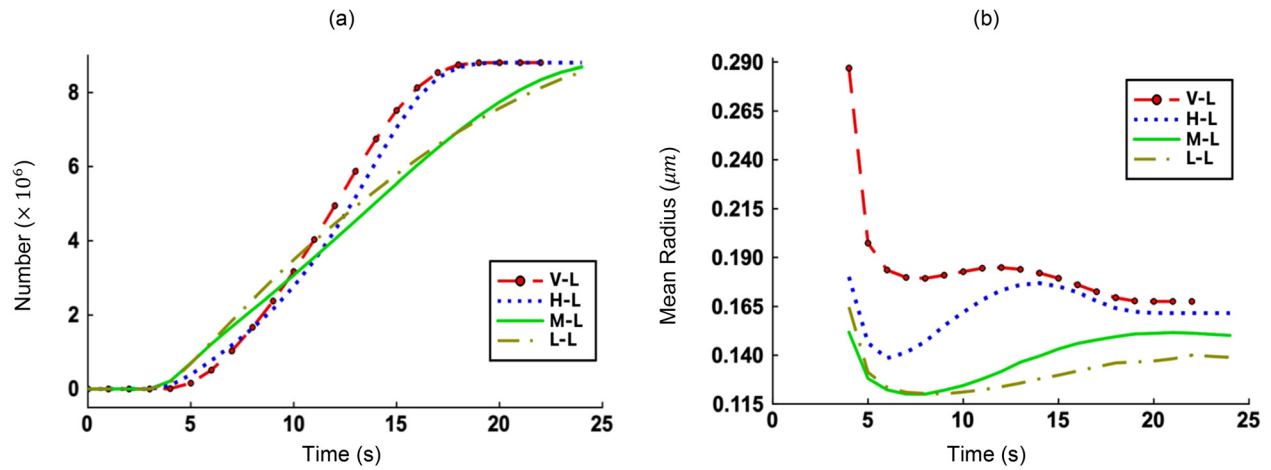


FIG. 19. Evolution of (a) number of aerosol particles, and (b) their mean radius. "V," "H," "M," and "L" before dash (–) stand for "very high," "high," "medium," and "low" turbulence levels.

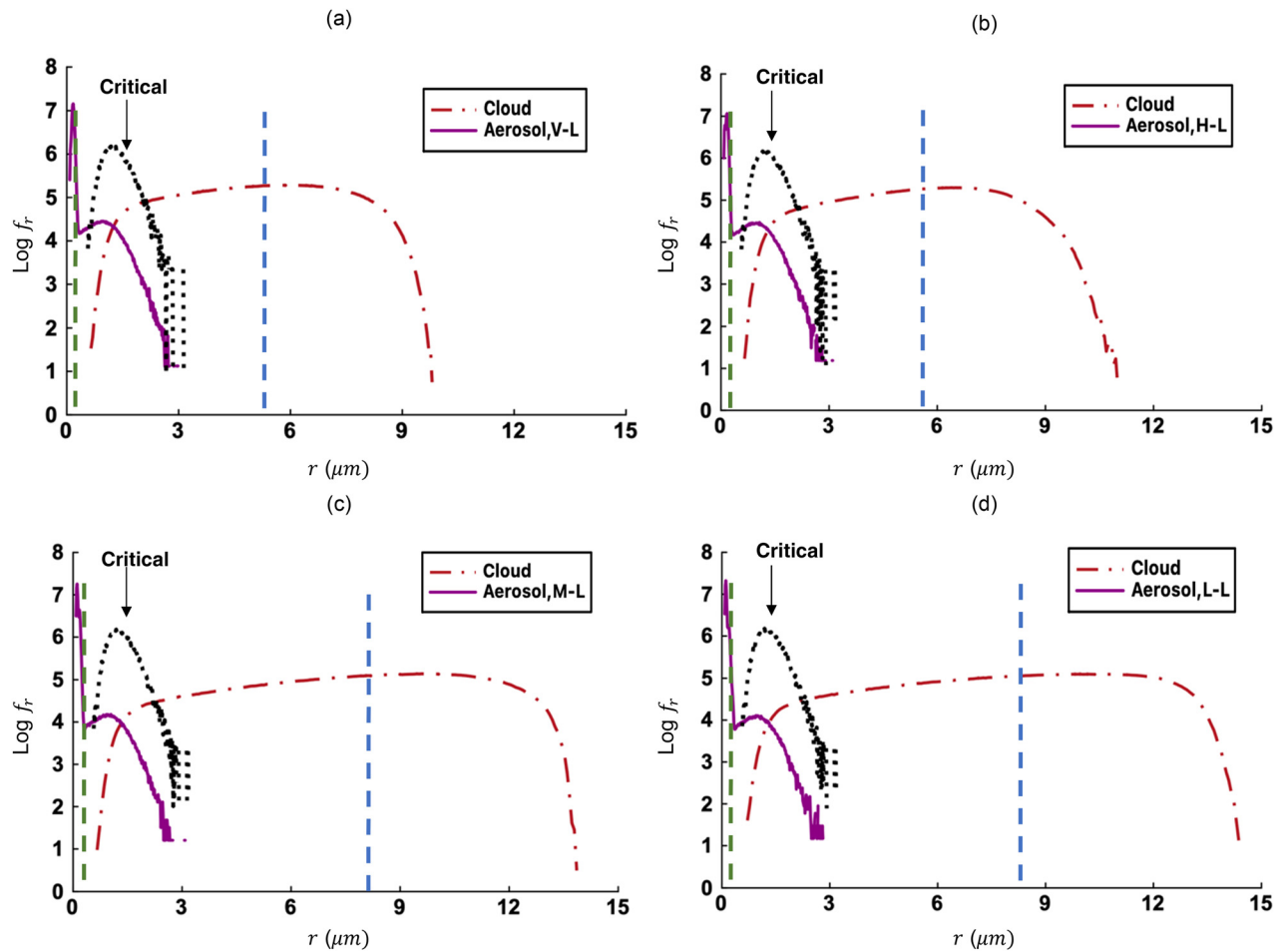


FIG. 20. Log PDF of particle size for cases (a) V-L, (b) H-L, (c) M-L, and (d) L-L, in scenario II at 12 s. "V," "H," "M," and "L" before dash (–) stand for "very high," "high," "medium," and "low" turbulence levels. The dotted line shows the critical radius (r_c). The green and blue dashed lines show the mean aerosol and cloud droplet radii, respectively.

larger in the aerosol population and increase the mean aerosol radii [Fig. 19(b)]. More discussion on effect of turbulence is deferred to Sec. V.

V. FURTHER DISCUSSIONS ON TURBULENT FLUCTUATIONS

This section describes the relationship between turbulent velocity and thermodynamic fields. The connection between fluctuations in the supersaturation and particle radius is also discussed.

The forcing of the velocity field is done to sustain flow turbulence and maintain finite amplitude of fluctuations in the otherwise zero-mean-velocity flow fields in homogeneous and isotropic turbulence. The time evolutions of the root mean square (rms) of the velocity fields (u_{rms}) are shown in Fig. 21(a) (scenario I) and Fig. 21(b) (scenario II) for different turbulence levels but the same (lognormal) dry aerosol size distribution. The u_{rms} goes through some transient period before reaching a statistically stationary state after approximately 6 s, with the details depending on the levels of turbulence.

Figures 22(a)–22(c) show the PDFs of fluctuating z-velocity, temperature, and vapor mixing ratio, respectively. These graphs represent scenario I at 10 s. The PDFs of fluctuating temperature and vapor mixing ratio in scenario II at 10 s are plotted in Figs. 22(d) and 22(e). We see in Fig. 22(a) that higher turbulence intensities broaden the distributions of the fluctuating z-velocity (w'), with significantly larger standard deviations from the mean, which is zero in this problem. The PDFs of the fluctuating x- and y-velocities are found to be similar, as expected from homogeneity and isotropy. However, the PDFs of the fluctuating thermodynamic fields are narrower for cases V–L and H–L compared to those for cases M–L and L–L in both scenario I [Figs. 22(b) and 22(c)] and scenario II [Figs. 22(d) and 22(e)]. This suggests that fluctuations, or deviations from the means, in the thermodynamic fields decrease with increasing flow turbulence intensity. Higher flow turbulence enhances mixing ($\rho\langle u'_i(x, t)\theta \rangle$), where θ is a placeholder for fluctuating temperature and water vapor mixing ratio, and the angular brackets, $\langle \cdot \rangle$, indicate spatial averaging. Due to stronger

mixing, passive thermodynamic fields⁴⁴ become more homogenized and fluctuations in these fields decrease.

To understand the connection between fluctuations in supersaturation and particle radius, we evaluate the quantity $\langle r'S'_e \rangle / \sigma_r \sigma_{S_e}$, which is the one-point correlation between r' and S'_e , normalized by the standard deviations. Figure 23 shows the time evolution of this quantity. We see that r' is strongly correlated with S'_e at earlier times, while the correlation decays over time. Moreover, the $\langle r'S'_e \rangle$ is always positive. Therefore, positive and negative fluctuations in S_e are associated with positive and negative fluctuations in r , respectively. Hence, any increase in the S'_e will broaden the spread of particle radii (σ_r).⁴⁵

A. Concluding remarks

The mutual interconversion of aerosol particles and cloud droplets in the atmosphere is numerically investigated by use of a finite difference-based direct numerical simulation (DNS) model. A spatially three-dimensional model in the core of atmospheric cloud is analyzed, and the combined effects of curvature and solute are investigated, as are the effects of turbulence intensity and the initial distribution of particles (monodisperse and polydisperse). The major results are summarized as follows:

- Air temperature increases while the vapor mixing ratio and supersaturation decrease faster when curvature and solute effects are included [Figs. 5(a)–5(c)], resulting in weaker condensation. The initial dry aerosol size distribution (monodisperse or lognormal) does not appear to affect the magnitude and temporal evolution of air temperature, supersaturation, and vapor mixing ratio [Figs. 5(a)–5(c)].
- For the same (high) turbulence intensity and monodisperse dry size distribution, activation is more rapid when curvature and solute effects are not included in the model [case H in Fig. 6(a)]. The mean particle radius is also larger for this case [Fig. 6(c)]. For the same (high) turbulence intensity, activation takes place more rapidly in our study when the initial size distribution of

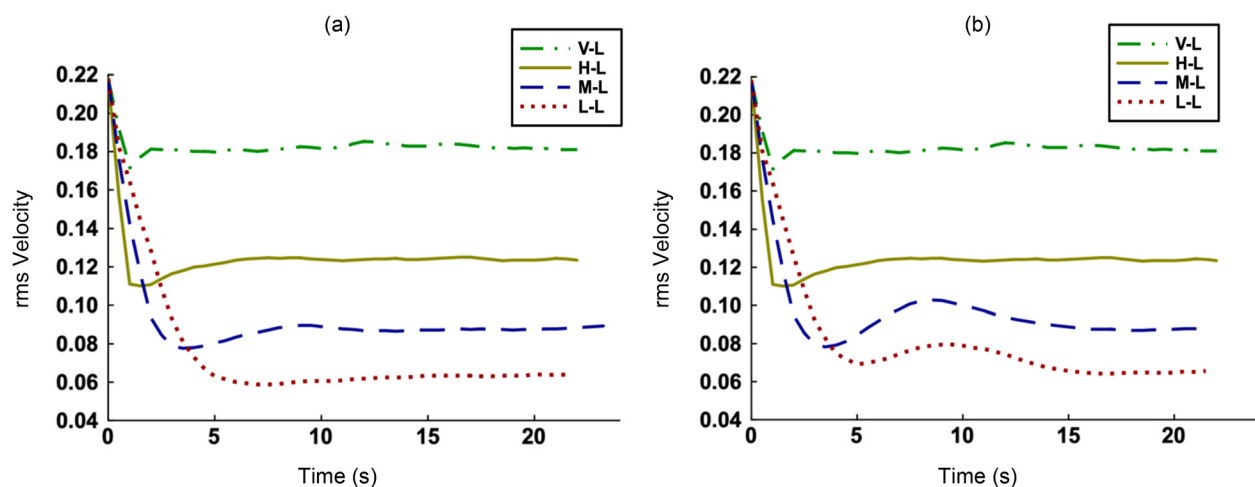


FIG. 21. Time evolution of the rms velocities in (a) scenario I and (b) scenario II. “V,” “H,” “M,” and “L” before dash (–) stand for “very high,” “high,” “medium,” and “low” turbulence levels.

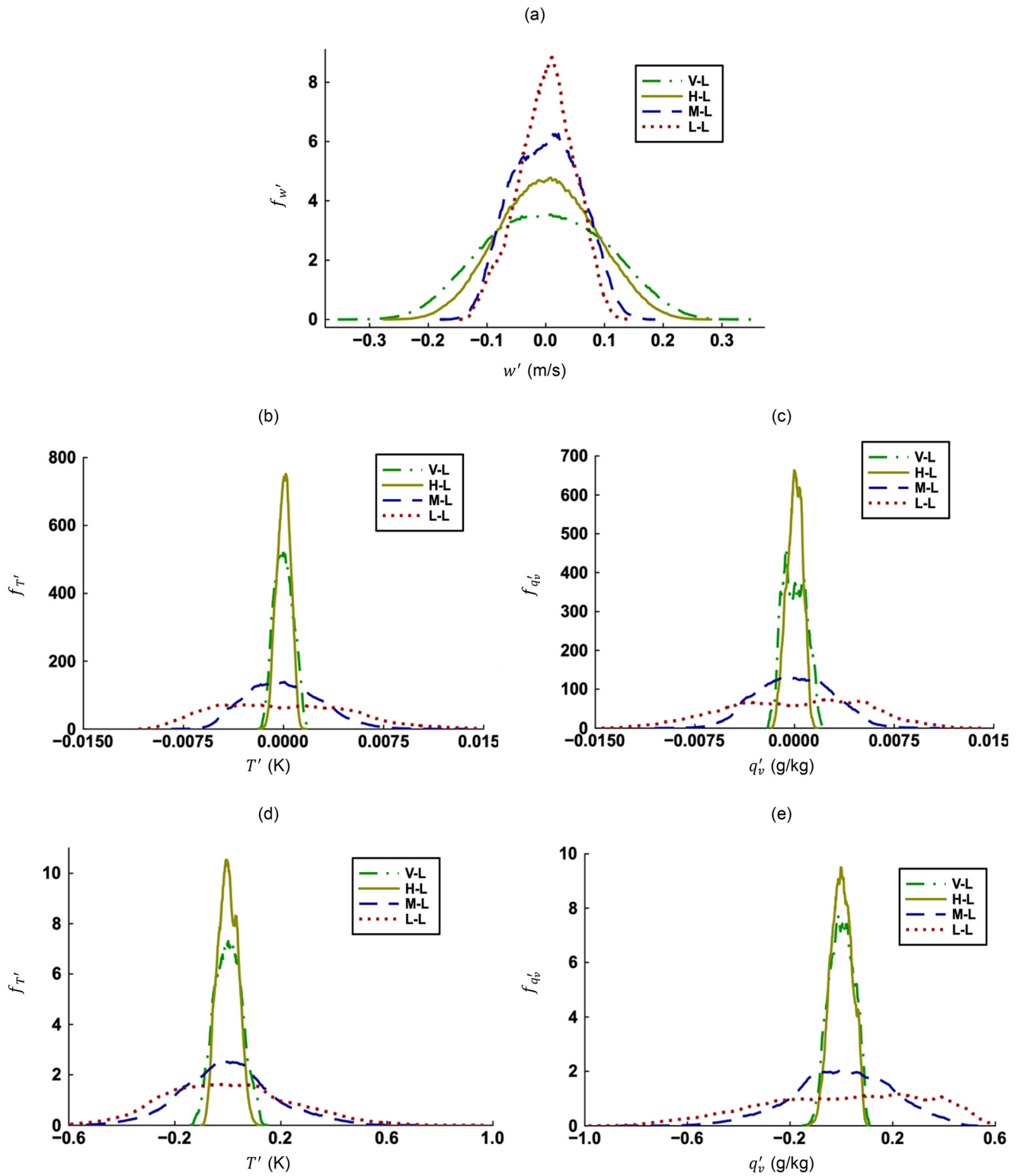


FIG. 22. PDFs of (a) fluctuating z-velocity (w'), (b) fluctuating temperature (T'), and (c) fluctuating vapor mixing ratio (q'_v), in scenario I at 10 s. PDFs of fluctuating (d) temperature and (e) vapor mixing ratio, in scenario II at 10 s. “V,” “H,” “M,” and “L” before dash (–) stand for “very high,” “high,” “medium,” and “low” turbulence levels.

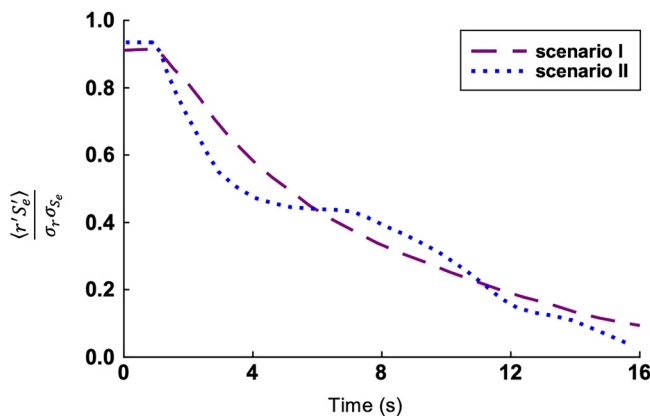


FIG. 23. Time evolution of the $\langle r'S_e' \rangle / \sigma_r \sigma_{S_e}$ for case L-L in scenarios I and II.

aerosol is monodisperse (case H-M) compared to lognormal (case H-L).

- With the lognormal initial dry size distribution, there is an overlap in the size distributions of aerosol particles and cloud droplets at short and medium time during simulation. That is, we can have a small particle classified as cloud droplet and a larger one as an aerosol particle. Also, larger aerosol particles in a lognormal distribution take more time to activate into cloud droplets compared to the smaller particles.
- For the cases of deactivation and high turbulence (H, H-M, H-L), neither the presence of dry aerosol (solute) nor its size distribution has any significant effects on air temperature, vapor mixing ratio, and supersaturation [Figs. 9(a)–9(c)]. The numbers of deactivated cloud droplets are also identical for the three cases [Fig. 10(a)].
- We have a temporal reduction in radius [Fig. 10(b)] as cloud droplets deactivate and evaporate. As the droplet radius decreases, curvature and solute effects start to become significant.
- Turbulence appears to delay activation [Fig. 15(a)]. The higher the turbulence intensity, the smaller the droplet radius [Fig. 15(b)] and the narrower the PDFs of droplet radius. But turbulence appears to promote deactivation [Fig. 19(a)] as the number of deactivated cloud droplets is higher with increasing turbulence intensity. The mean radius of aerosol particles (from deactivated cloud droplets) is larger when turbulence intensity is higher [Fig. 19(b)].
- Activation appears to be mean-dominated in scenario I (Fig. 14), although with possible fluctuation-influenced activation at early time. Deactivation is mean-dominated in scenario II (Fig. 18), but with possibility of fluctuation-influenced deactivation. Prabhakaran *et al.*¹² reported on mean-dominated and fluctuation-influenced regimes from simplified one-dimensional theoretical treatment and physical experiments. Our approach is based on full three-dimensional DNS modeling.
- Some previous models^{46,47} which studied aerosol-cloud interactions assumed a homogeneous distribution in the initial thermodynamic fields. But our initial configuration includes an interface across which the thermodynamic fields change sharply. This realistic setup allows us to evaluate the effects of turbulent mixing on activation and deactivation thoroughly.

- Stronger turbulence leads to increased homogenization of the passive thermodynamic fields [Figs. 22(b)–22(e)] in our study. Hence, the smaller variance of environmental supersaturation is associated with higher turbulence intensities [Figs. 13(b) and 17(b)]. Vaillancourt *et al.*⁴⁸ also reported that fluctuations in supersaturation decrease when the turbulence intensity increases, even though their domain length in each spatial coordinate direction is one-fifth of the one used in the present study (0.10 vs 0.512 m).
- The radius PDFs of aerosol particles are found to be bimodal during deactivation (Fig. 12). Fluctuations in supersaturation broaden the spread of cloud droplet radii (Fig. 23). The broadening is followed by a narrowing (Figs. 7, 8, 11, and 12) because turbulent homogenization reduces thermodynamic fluctuations over time.
- Large-scale atmospheric models parameterize the effects of turbulence and microphysics instead of explicitly resolving them as in this DNS study. Under the various conditions investigated, the large-scale models depend on the microscale results such as those in the present investigation to provide the subgrid-scale models. For example, the results from our DNS study can provide the required subgrid-scale information for large eddy simulations (LES), which could then be applied to even larger-scale models.

It is noteworthy that the present work on activation and deactivation can be improved in the future by considering the following points.

- We have focused on the influence of turbulence through external forcing in the momentum field only, with the impact of external thermodynamic forcing being ignored. Our future work will consider external thermodynamic forcing, wherein the competition between turbulent homogenization and source of thermodynamic variance is expected.
- Particle-based DNS is a relatively new development in the cloud physics community. These models usually have a spatial grid resolution of millimeters whereas typical measurements from laboratory and aircraft-based experimental campaigns have a lower resolution (approximately 100 m). The latest development in the digital holography⁴⁹ is changing the situation. It will be desirable to conduct direct comparisons of DNS and observational data when the latter becomes available.
- The dry aerosol number concentration in our model is $\sim 65 \text{ cm}^{-3}$, corresponding to a clean environment. But the number concentration can reach $\sim 10^4 \text{ cm}^{-3}$ in a polluted environment. That corresponds to $\sim 10^{10}$ particles for our domain size. Although a comparison of the results for clean and polluted environments would be interesting, the latter is computationally prohibitive for a particle-based DNS approach. The present study (8.8×10^6 particles, 0.512^3 m^3 volume) tracks more particles than the DNS model of Thomas *et al.*¹¹ (3.9×10^5 particles, 0.128^3 m^3 volume) which simulated aerosol activation. Recently, Dennis *et al.*⁵⁰ simulated 10^9 Lagrangian particles but without explicitly resolving small-scale turbulence and using LES instead. With the increase in computational power, DNS studies of activation and deactivation in a polluted environment ($\sim 10^{10}$ particles) will be attempted in the future.
- We run simulations for one type of dry aerosol (ammonium sulfate). In the future, the chemical composition of aerosol and thus

hygroscopicity¹⁵ can be varied to enhance investigation into the influence of chemical composition on activation and deactivation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Abdullah Al Muti Sharfuddin: Formal analysis (equal); Methodology (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Foluso Ladeinde:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Yangang Liu:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Mohammad Atif:** Data curation (equal); Formal

analysis (equal); Methodology (equal); Software (equal); Validation (equal). **Fan Yang:** Formal analysis (equal); Investigation (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Vanessa Lopez-Marrero:** Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Validation (equal). **Meifeng Lin:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Software (equal); Validation (equal). **Tao Zhang:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: SOME DETAILS ON TURBULENCE FORCING

To create the “low,” “medium,” “high,” and “very high” turbulence levels, we have applied forcing in the following wavenumber bands: $1 \leq k_f/k_{min} \leq 2.45$, $1 \leq k_f/k_{min} \leq 4.12$, $1 \leq k_f/k_{min} \leq 8.12$, and $1 \leq k_f/k_{min} \leq 17.32$, respectively, where k_f is the forced wavenumber, and $k_{min} = 2\pi/L = 12.27$ is the minimum wavenumber using $L = 0.512$ m. The maximum wavenumber is $k_{max} = N\pi/L = 1570.8$ using $N = 256$. Note that $k_{max} \sim 128k_{min}$. The ratios of k_{max} to the largest k_f are 52.24, 31.07, 15.76, and 7.39, respectively. Eswaran and Pope¹⁸ used $2\sqrt{2}k_{min}$ as the largest k_f in a 64^3 simulation, with $k_{max} \sim 30.17k_{min}$ and a ratio of k_{max} to the largest k_f of 10.67. Therefore, the forced wavenumber bands in our model are narrower than that in Ref. 18 except for the case of very high turbulence level. According to Eq. (4), variation of either ϵ_{in} or the wavenumber band should produce different turbulence intensities. However we have observed that ϵ_{in} does not have a significant impact in this regard. We ran four test cases that differ in either ϵ_{in} , the wavenumber band, or both. The ϵ_{in} was varied from 0.004 to 0.016 m^2/s^3 , while the wavenumber band was varied from $1 \leq k_f/k_{min} \leq 4.12$ to $1 \leq k_f/k_{min} \leq 8.12$. Figure 24(a) shows that the TKE dissipation rate

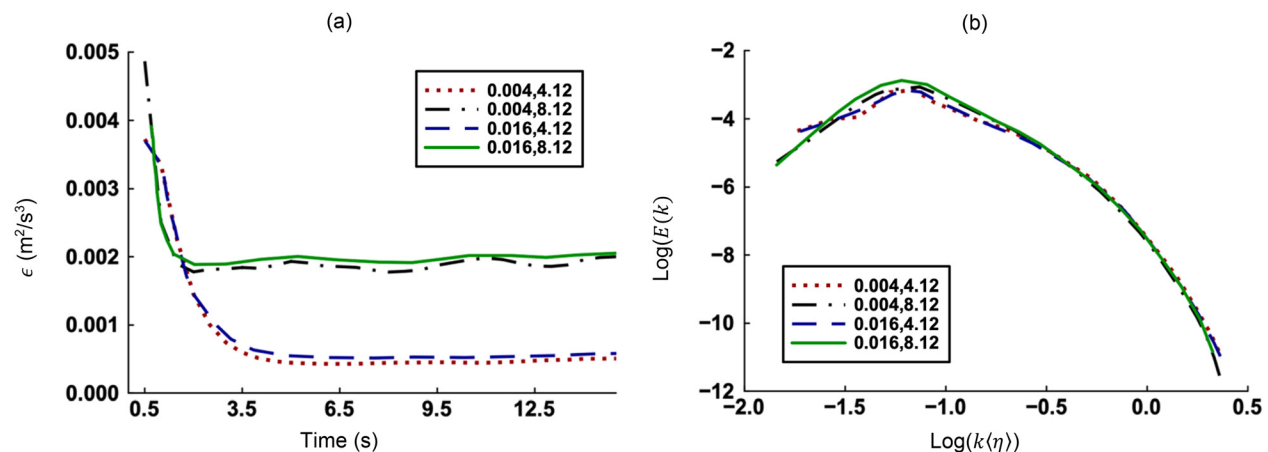


FIG. 24. (a) Time evolution of TKE dissipation rate (ϵ), and (b) TKE spectra at 6 s in log format, for the four cases. Note that wavenumbers (k) in the abscissa are multiples of k_{min} , and are in the radial coordinate direction. In the legends, the value of ϵ_{in} (m^2/s^3) and the maximum forcing wavenumber are separated by a comma.

increases only marginally with increasing ϵ_{in} , but significantly with changes in the wavenumber band. Figure 24(b) suggests that the four spectra converge at intermediate and small scales while being different at large scales, as expected. Hence, the different results discussed in Sec. IV B can be attributed to differences in k_f .

The initial velocity field has been constructed in the Fourier space⁵¹ to satisfy continuity, isotropy, and homogeneity with a given energy spectrum. A regular cubic grid of points, each of which supports a single Fourier mode,¹⁸ is employed in the Fourier space. The initial TKE spectrum from the velocity is specified as⁵²

$$E(k) = \frac{16}{\sqrt{\pi/2}} \frac{u_0^2 k^4}{k_0^5} \exp\left(-\frac{2k^2}{k_0^2}\right), \quad (A1)$$

where $k_0 = 2.45k_{min}$ and $u_0 = 0.22$ m/s is the rms velocity from the generated fluctuating velocity field. The form k^4 for low wavenumbers correspond to the incompressible limit while the exponential function allows a rapid roll-off of the energy at high wavenumbers.

Of interest in our analysis is the integral length scale, l , which is calculated as⁵³

$$l = \int_0^\infty R_{w'w'}(r) dr. \quad (A2)$$

The autocorrelation coefficient, $R_{w'w'}$, is defined as⁴⁴

$$R_{w'w'}(r) = \frac{\langle w'(\mathbf{x}, t) w'(\mathbf{x} + \mathbf{r}, t) \rangle}{\sqrt{\langle (w'(\mathbf{x}, t))^2 \rangle \langle (w'(\mathbf{x} + \mathbf{r}, t))^2 \rangle}}, \quad \mathbf{r} = \Delta x \mathbf{i} + \Delta y \mathbf{j} + \Delta z \mathbf{k}, \quad (A3)$$

where w' is the fluctuating z-velocity and $\mathbf{r}$ is the separation distance vector. The l can be interpreted as the separation distance in Fig. 25 at which the autocorrelation is approximately zero. The values of l using Fig. 25 are 0.18, 0.15, 0.12, and 0.10 m, respectively for “low,” “medium,” “high,” and “very high” turbulence level cases. With a domain length (L) of 0.512 m, the corresponding values of L/l are 2.84, 3.41, 4.27, and 5.12. We are aware of suggestions that the

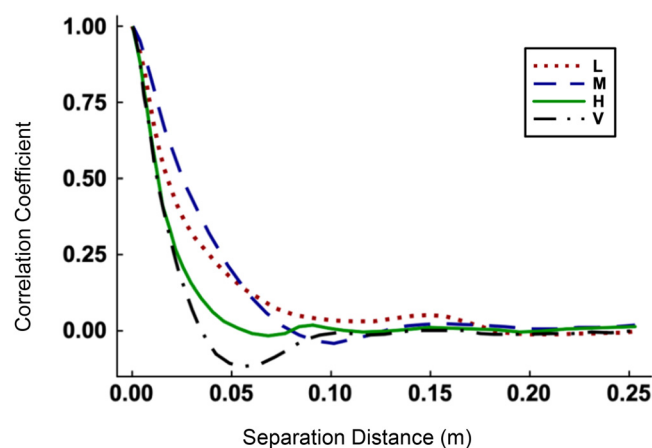


FIG. 25. The autocorrelation coefficients for z-velocity at 6 s. “L,” “M,” “H,” and “V” stand for “low,” “medium,” “high,” and “very high” turbulence levels.

typical l should be one-eighth⁵¹ of the domain size (L). However, these are based on scaling arguments and not exact relationships. The box length is sufficiently larger than the integral length scales for our purpose. Note that previous studies by Pope and his co-workers did not use an L/l of 8 but rather assumed values of approximately 4.5 (Ref. 54) and 3.0.¹⁸

APPENDIX B: ASSUMPTIONS IN THE PRESENT MODEL

The assumptions of this study are itemized below:

- (1) Flow turbulence is homogeneous and isotropic. The velocity is divergence-free, and the flow is incompressible. The mean velocity fields ($\langle u \rangle$, $\langle v \rangle$, and $\langle w \rangle$) are zero.
- (2) A cloud droplet is in a state of equilibrium with surrounding water vapor, and thus, the Kappa-Kohler theory is applicable.
- (3) The thermodynamic fields are passive and do not influence the momentum field. However, the former are significantly impacted by the latter.
- (4) The Stokes number is significantly less than unity, and thus, particles are freely carried around by the continuous air phase.
- (5) Particles are impacted by the drag force and gravity only. The relative acceleration between the particle and fluid is negligible and so is the Basset history force.
- (6) Particles do not interact with each other, do not merge or deform, and are perfectly spherical.
- (7) The model's domain length is much smaller compared to an actual cloud.

APPENDIX C: HEIGHT-DEPENDENCY OF THE MEAN QUANTITIES

The mean temperature, pressure, and density vary with height (elevation) in the real atmosphere and in large-scale atmospheric models. However, such variation is negligible in our microscale model. We can write

$$T_2 = T_0 - Bz, \quad p_2 = p_0 \left(1 - \frac{Bz}{T_0}\right)^{\frac{\gamma}{\gamma-1}}, \quad \rho_a = \frac{p}{R_d T}. \quad (C1)$$

For the present study, $z = 2677.44$ m is the distance between model's top and sea level, $R_d = 287$ J/(kgK), and $B = 0.0065$ K/m is the adiabatic lapse rate, with $T_0 = 288.15$ K, $p_0 = 101325$ N/m², and $\rho_{a,0} = p_0/(R_d T_0) = 1.225$ kg/m³ as the datum. Using Eq. (C1), the corresponding values at the bottom of our model are $T_1 = 270.75$ K, $p_1 = 71446$ N/m², and $\rho_{a,1} = 0.92$ kg/m³ while those at the top are $T_2 = 270.746$ K, $p_2 = 71440$ N/m², and $\rho_{a,2} = 0.92$ kg/m³. Hence, it is evident that T , p , and ρ do not change significantly with height in the domain of our microscale model, unlike in models with significantly larger domains (mesoscale and global models). Note that the density change is directly related to the temperature change as follows:

$$\rho_a = \rho_{a,0}(1 + \beta \Delta T), \quad \beta = \frac{1}{\rho_a} \left. \frac{\partial \rho_a}{\partial T} \right|_{T_0}, \quad (C2)$$

where β is the coefficient of thermal expansion. Again, our model does not account for the work of expansion and density change

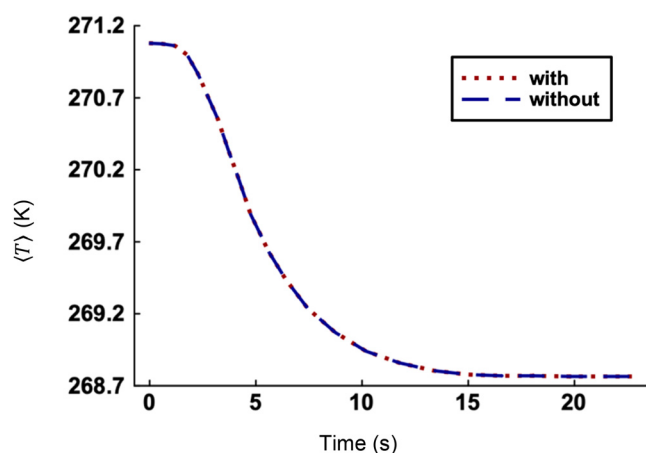


FIG. 26. Time evolution of the mean temperature ($\langle T \rangle$) with and without adiabatic lapse rate for case L-L in scenario II.

with temperature relative to the base density is assumed to be negligible. To verify this, we carried out simulations that included the adiabatic lapse rate term⁷ ($-g/c_p$) w in Eq. (5) but found no influence on the results (Fig. 26). This will be different for mesoscale and global atmospheric models.

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